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Design of a Racing Bicycle using a Computational Software Package

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ABSTRACT

The objective of this project is to design a racing bicycle frame using computational software. This report outlines the methodology, analysis, and findings of the design process. By leveraging advanced computational tools, the aim is to optimize the frame geometry for enhanced performance and efficiency. Furthermore, the abstract explores the cutting-edge of racing bicycle design, using contemporary computer software to optimize the frame's performance and economy. In the current competitive cycling age, when aerodynamic supremacy and precise engineering are critical, the use of computer technologies presents a paradigm change in the design process. This research uses sophisticated optimization methods, computational fluid dynamics (CFD), and finite element analysis (FEA) to transform established design processes through rigorous analysis and simulation. The main goal is to create a racing bicycle frame with an unmatched stiffness-to-weight ratio, aerodynamic brilliance, and rider comfort that not only matches but exceeds the demanding requirements of elite riders.

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NOMENCLATURE

FEA	Finite Element Analysis
CFD	Computational Fluid Dynamics
CAD	Computer-Aided Design
CFRP	Carbon Fiber Reinforced Polymer
HSLR	High Seat Low Rear
SLA	Selective Laser Annealing
SUC	Situated Use Cases
XCO	Olympic Cross-country
XCM	Cross-country marathon
XCP	Cross-country point-point
XCE	Cross-country short track
XCT	Cross-country eliminator
XCR	Cross-country time trial
XCS	Cross-country stage race

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1. INTRODUCTION

1.1. Executive Summary

Engineers and designers work to improve their tools of the trade, the bicycle frame, to optimize speed, agility, and efficiency while complimenting the athletes' dedication to exceed the current accolades. For a considerable amount of time, traditional bicycle design techniques have depended on a blend of expertise, intuition, and empirical testing to obtain important performance metrics for continuous improvement. The design process for bicycles is iterative as it involves constant refinement due to technological advancements, user feedback, performance optimization based on the athlete's performance metrics to meet the demands and adapt to the current trends in the world of cycling. Numerous design possibilities have opened with the development of sophisticated modeling tools and computer software to simulate virtual prototypes for maximum cost efficiency while also yielding accuracy. These bicycle prototypes can be simulated in accurate test environments through the use of mathematical models and programming to simulate the performance by measuring metrics such as stress, displacement, drag, velocity.

This introduction lays the groundwork for investigating the nexus between technology and cycling, where the combination of digital accuracy and human creativity promises to push the limits of what is feasible in the design of racing bicycles. We explore the complexities of computational design approaches and consider how they can transform the production of racing bicycles.

1.2. Sport Background

Competitive bicycle racing comprises many events, ranging from short-distance sprints to road races covering distances from 30 to over 5000 km, as well as mountain bike, BMX and para-cycling competitions [10]. Road racing contains various types of competitions, including traditional road races, circuit races, criteriums, time trials, hill climbs, and stage races. All road races are mass start events with competitors separated into categories based on gender, skill, and/or age groups [11]. Differences between bicycles marketed for road, gravel, and cyclocross

disciplines are mainly focused on tire clearance and frame geometry. Traditionally, road racing bicycles feature tight tolerances for the tire clearance and angles suited to high-speed cornering and stability for descents [11]. Bicycles designed for road, gravel and cyclocross can all be used for road racing, and there are no bike restrictions on tire width for road cycling. However, the differences in handling characteristics will influence rider confidences and comfort [11]

1.3. General Approach to the Problem/Design

An in-depth understanding of the demands and limitations placed on the racing cycling environment is the first step in addressing the design problem of developing an optimum racing bicycle frame. Initial stages will be spent on conducting research to understand the bicycle frame and its corresponding components. In addition, research will be conducted to understand the simulation methods to evaluate the performance of a bicycle. Adequate research is critical to understand the design processes that were used by experts and use that as a basis for our design process of the racing bicycle.

This results in the examination of cost-effectiveness and production feasibility coupled with performance criteria including weight, stiffness, aerodynamics, and rider comfort. The design process is iterative, based on the most recent developments in computational tools and informed by a foundation of empirical information and industry best practices. In the early stages of idea generation, a variety of design options are explored through brainstorming meetings and drawing activities. Then, using computational simulations and analysis, these ideas are rigorously examined. FEA is used to evaluate structural integrity, magnitude of von-mises stress and deflection.

In summary, the following literature review provides in-depth analysis as to the meticulous balance of engineering precision, material selection, and aerodynamic analysis. Every aspect of the bike, from its frame to its transmission systems, plays a crucial role in determining its performance on the track. In this section, the best practices are to be employed by bike designers and engineers to create top-tier racing bicycles that excel in speed, agility, and efficiency. Moreover, the following table outlines key best practices in three critical areas of racing bicycle design:

Table 1: Best Practices for designing each component for the bicycle.

Aspect	Best Practices
Frame Design	➤ utilize lightweight materials such as carbon fiber or aluminum for optimal strength-to-weight ratio. ➤ focus on aerodynamics to reduce drag in certain areas such as downtube, seat tube, and seat. ➤ incorporate stiff geometries for efficient power transfer
Tire Analysis	➤ tire characteristics are important for bicycle dynamics ➤ optimize tire width and tread pattern for specific racing conditions (eg. road surface, weather, etc.). ➤ consider choosing tires with low rolling resistance to minimize friction and increase speed.
Transmission Systems	➤ implement a lightweight and efficient drivetrain, typically with an efficient chain, cassette and crankset to increase bicycle performance. ➤ select gearing ratio suitable for race terrain, balancing climbing and plat speed. ➤ ensure smooth shifting mechanisms to minimize energy loss during gear shifts.

1.4. Literature Review

Bicycle frames represent the foundational structure of every bicycle, serving as the backbone that defines its performance, durability, and ride quality. Over the years, materials and frame design used in bicycle frames have undergone significant evolution due to advancement in material processing, engineering and demand for lighter, stiffer bicycles. Advancements in computation modeling, finite element analysis have revolutionized the design and manufacturing processes of bike frames, enabling engineers to push the boundaries of performance while ensuring structural integrity and safety standards.

Initial research was conducted to study the bicycle frame components and designs that are readily available in the market. Figures 1-3 show the common types of bicycle frame design with varying cross sections. Typical bicycle frame consists of several key components such as the Top Tube, Head Tube, Seat Tube, Bottom Tube, Chain Stays, Seat Stays, which are joined together using welding processes.



Figure 1: High Seated Diamond Frame [12]



Figure 2: Low Seated Diamond Frame [13]



Figure 3: HSLR Diamond Frame [14]

Stress analysis using computational simulations are beneficial to determine the particular zones where stress and displacement is high and allow engineers to further improve the design. Stress analysis of a bicycle frame for various materials was performed by Sarath P et al using FEA (Finite Element Analysis) to understand the behavior of the bicycle frame for various materials and load conditions [5]. The design parameters of the bicycle frame are shown in Table 1, with the sketched and 3D model shown in Figure 4.

Table 2: Design Parameters of the Bicycle Frame [5]

Parameter	Value
Seat Tube Angle	73.5°
Seat Tube Length	580 mm
Top Tube Length	570 mm

Bottom Tube Length	670 mm
Chain Stay Length	360 mm
Seat Stay Length	500 mm



Figure 4: 3D model of the bicycle frame with labeled parameters [5]

According to the Sarath P, et al, stress analysis of the bicycle frame was performed using ANSYS undergoing 5 different types of loading conditions: Static start-up. Steady State Pedaling, Vertical Impact, Horizontal impact and Rear Wheel braking [5]. The five loading conditions are summarized in Table 2.

Table 3: Loading Conditions performed on the bicycle frame [5]

Loading Condition	Description
Static Start-up	Load of 700 N and 200 N applied to the top of the seat tube and bottom of bracket respectively
Steady State Pedaling	Load of 1000N and 200 N applied to the bottom tube and head tube respectively
Vertical Impact	Load of 2250 N applied to the top of the bracket
Horizontal Impact	Load of 2250 N applied to the head tube horizontally
Rear Wheel Braking	Load of 750 N applied to the rear wheel bracket

According to the simulations performed by Sarath P, et al, the maximum equivalent stress and maximum deformation of the bicycle frame undergoing the 5 loading conditions were tested on ANSYS and summarized in Tables 3, 4.

Table 4: Comparison of Maximum Equivalent (Von-Mises) Stress (MPa) for different materials [5]

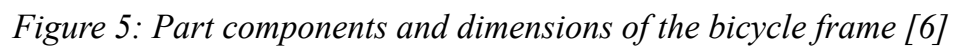
	Maximum Equivalent (Von-Mises) Stress (MPa)			
Conditions	Steel	Aluminum 6061 T6	Titanium Grade 9	Carbon Fiber
Static Startup	9.8212	10.915	10.896	10.940
Steady State Pedaling	6.3773	7.4270	7.3168	8.4552
Vertical Impact	31.216	33.722	34.607	34.839
Horizontal Impact	28.069	30.364	31.071	33.941
Rear Wheel Braking	17.746	17.780	18.022	19.498

Table 5: Comparison of Maximum Deformation (mm) for different materials [5]

	Maximum Total Deformation (mm)			
Conditions	Steel	Aluminum 6061 T6	Titanium Grade 9	Carbon Fiber
Static Startup	0.0081404	0.023512	0.017838	0.0040985
Steady State Pedaling	0.0084963	0.024481	0.018578	0.0042668
Vertical Impact	0.0024997	0.0072276	0.0054793	0.012585
Horizontal Impact	0.030902	0.090021	0.068393	0.015749
Rear Wheel Braking	0.18428	0.52604	0.39793	0.089395

From Table 3, the carbon fiber had the largest Maximum Equivalent (von Mises) Stress, followed by Aluminum 6061 T6, Titanium Grade 9, and then Steel. The carbon fiber bicycle frame also had the smallest value of maximum total deformation, followed by Aluminum 6061

Another study was performed by Akhyar et al. to study the bicycle frame behavior under various load conditions for a T-profile and I-profile cross section using Aluminum 6061 T6 which are shown in Figures 5, 6 [6]. The load conditions performed by Akhyar et al. were identical to the simulation performed by Sarath P, et al, which were analyzed and compared to understand how the differences in bicycle frame geometry can affect the maximum deformation and maximum equivalent stress values at critical locations.



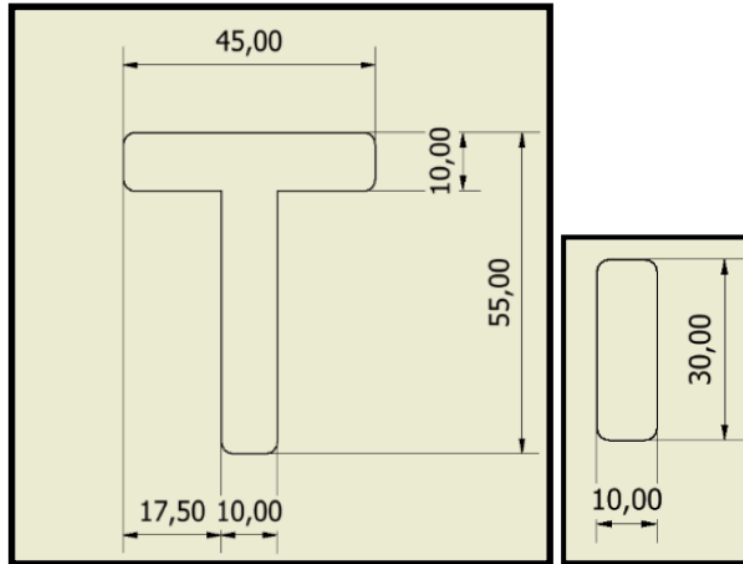


Figure 6: Dimensions of the T-Profile and I-Profile Cross Sections [6]

The bicycle frame with the T-Profile, I-Profile Cross sections has been manufactured using gravity sand casting technique as shown in Figure 7 [6]. According to Akhyar et al., bicycle frames that are procured from metal pipe assembly through the use of welding technology, have problems with material degradation in welding and Heat Affected Zone (HAZ) area [6]. Although no welding processes are used in the design, a bolted connection between the main frame and seat stay was still required [5].



Figure 7: Gravity Sand Casting processes to manufacture the bicycle frame [6]

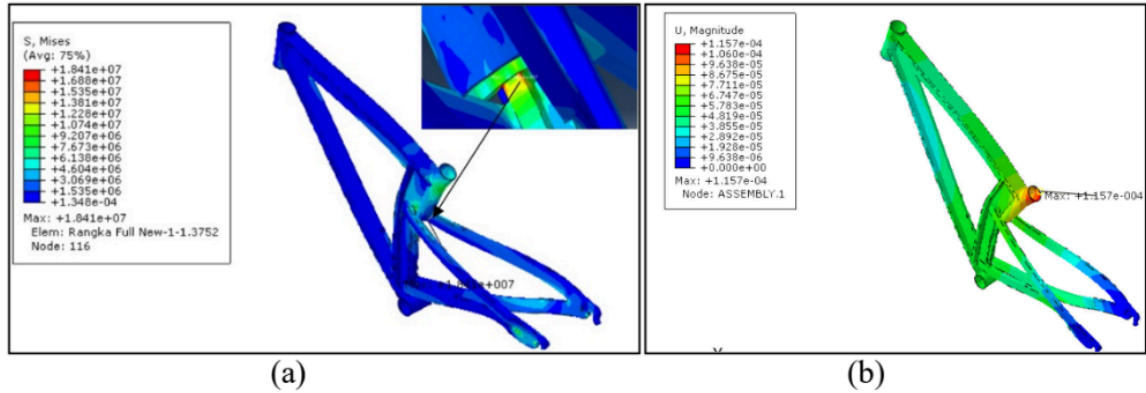


Figure 8: FEA simulation for Static start-up loading condition (a) Von-mises stresses (MPa) (b) Displacement [6]

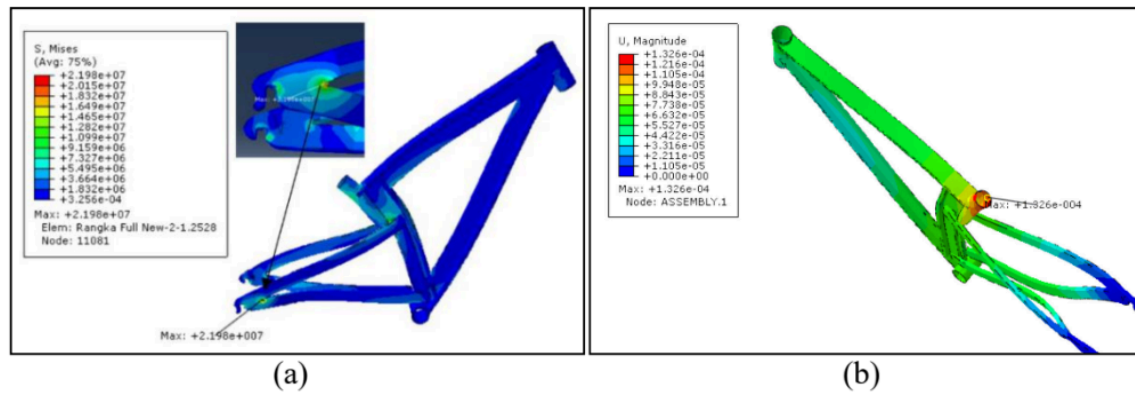


Figure 9: FEA simulation for Steady-state pedaling loading condition (a) Von-mises stresses (MPa) (b) Displacement [6]

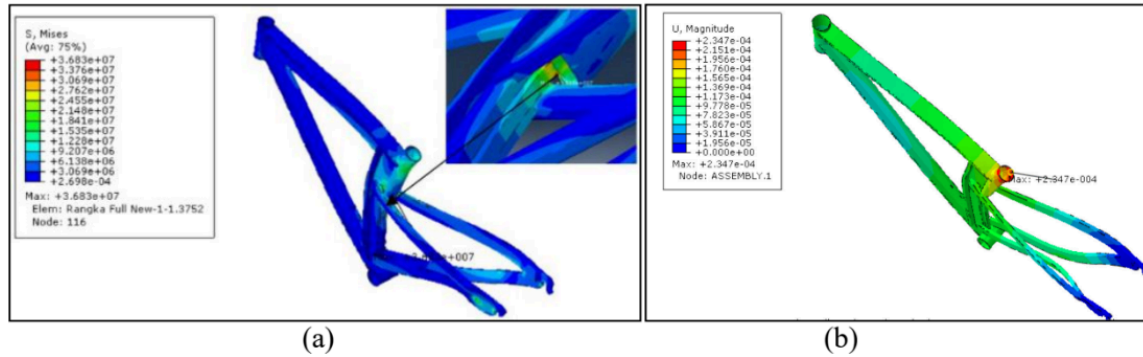


Figure 10: FEA simulation for Vertical Loading loading condition (a) Von-mises stresses (MPa) (b) Displacement [6]

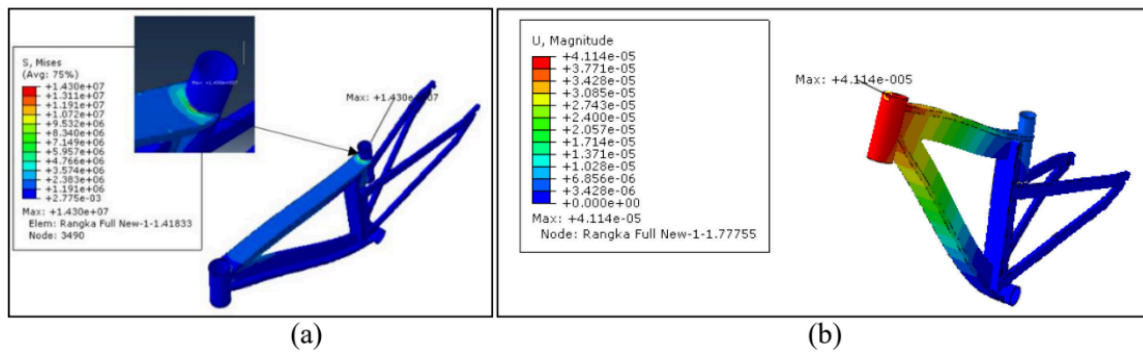


Figure 11: FEA simulation for Horizontal loading condition (a) Von-mises stresses (MPa) (b) Displacement [6]

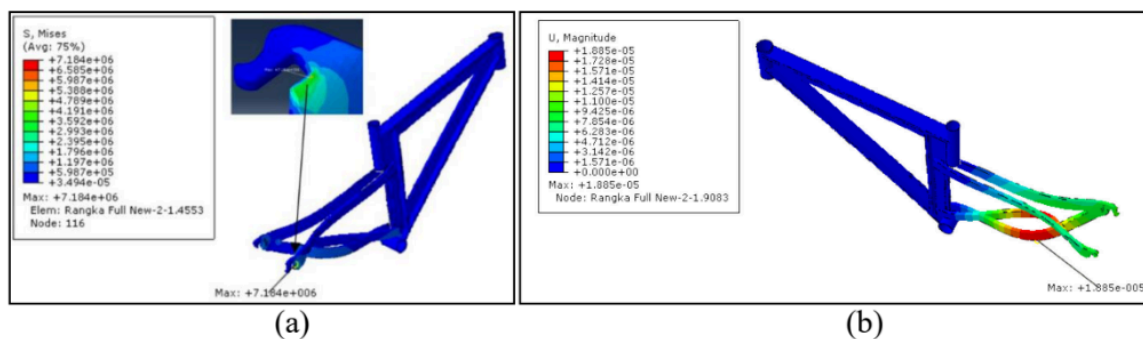


Figure 12: FEA simulation for Rear Braking loading condition (a) Von-mises stresses (MPa) (b) Displacement [6]

Figures 8 to 12 are the FEA simulation results obtained by Akhyar et al. for the T-Profile and I-Profile bicycle frame for the 5 loading conditions. Image (a) shows the magnitude of Von-mises stresses (MPa) and image (b) shows the displacement (mm). Note that the red areas for all 5 loading conditions are the critical areas where displacement values were largest.

Table 6: Summary of Maximum von-mises Stress (MPa) and Displacement (mm) [6]

Location	Static start-up		Steady-state pedaling		Vertical loading		Horizontal loading		Rear wheel braking	
	stress	dis.	stress	dis.	stress	dis.	stress	dis.	stress	dis.
Seat beam	18.4	-	-	-	36.83	-	-	-	-	-
Chain stay-seat stay	-	-	21.98	-	-	-	-	-	-	-
Top beam	-	-	-	-	-	-	14.3	-	-	-
Drop out	-	-	-	-	-	-	-	-	7.18	0.018
Seat stay	-	0.11	-	0.13	-	-	-	-	-	-
Seat tube	-	-	-	-	-	0.23	-	-	-	-
Head tube	-	-	-	-	-	-	-	0.041	-	-

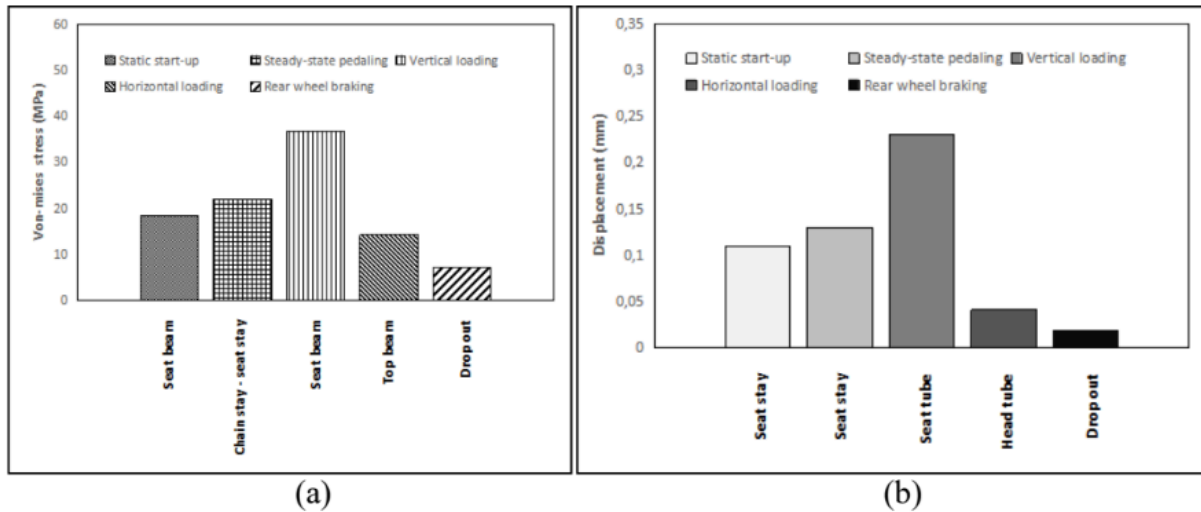


Figure 13: (a) Von-Mises Stress (MPa) and (b) Displacement (mm) for the different loading conditions on the bicycle frame [6]

The summary of the FEA simulations performed by Akhyar et al. are shown in Table 5 and graphically represented in Figure 13. The Von-Mises Stresses ranged from 7.18 - 36.83 MPa and Displacements ranged from 0.018 - 0.23 mm. The simulation results for Aluminum 6061 T6 performed by Akhyar et al. and Sarath P, et al yield similar results but there was a significant

improvements in the stress and displacement values in the Rear Wheel braking loading condition for the T-Profile and I-Profile Cross section bicycle frame.

Another study was performed by Alexandros G. Kitselis et al, to recommend an alternative design process of an optimized road racing bicycle frame. The optimization of the bicycle frame was carried out using CFD and CAE iterations to evaluate the part performances [6]. The design process started with a basic tube diagram of a typical bike frame as shown in Figure 5 and after 9 bicycle frame iterations, the optimized bike frame design is finalized as shown in Figure 12.

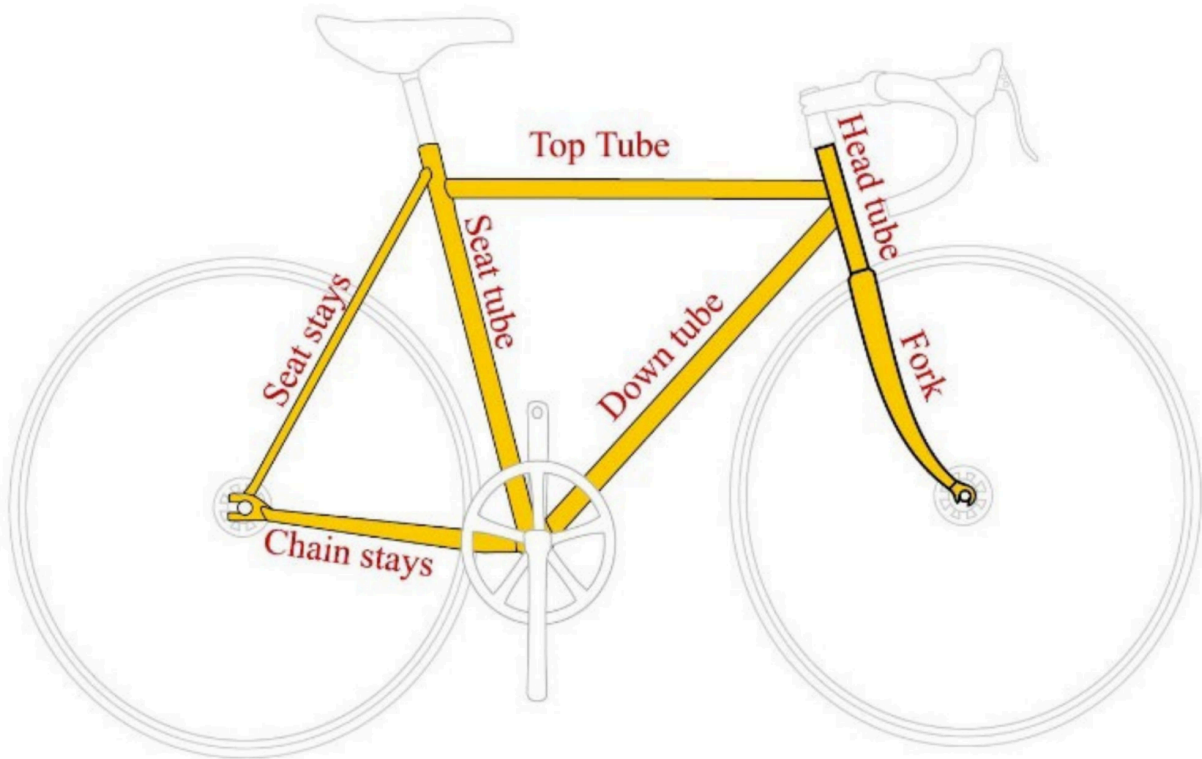


Figure 14: Basic tube diagram of a typical frame [4]

Figure 14 depicts a diagram of a bicycle, highlighting and labeling various parts of its frame:

Different parts of the bicycle are labeled:

- “Top Tube” is written near the upper horizontal bar.
- “Head tube” is indicated at the front vertical bar connecting the handlebars to the front wheel.

- “Fork” is labeled on the part connecting the head tube to the front wheel.
- “Down tube” is written next to the diagonal bar connecting the head tube to the pedal area.
- “Seat tube” labels the vertical bar supporting the seat.
- “Seat stays” are indicated on bars connecting the seat area to the rear wheel.
- “Chain stays” label is placed near the lower bars connecting the pedal area to the rear wheel.

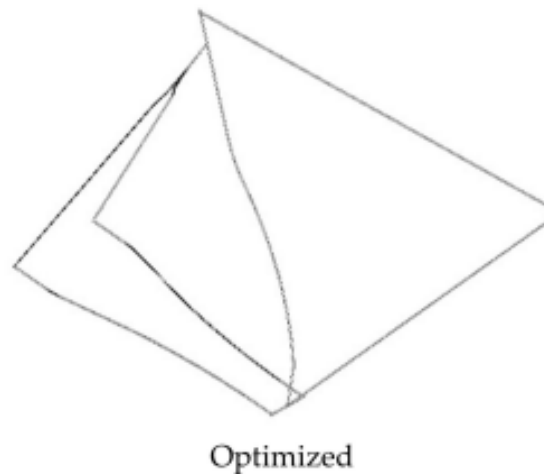


Figure 15: Optimized Bicycle Frame [6]

According to Alexandros G. Kitselis et al, the testing methods of the bicycle frame were based on the ISO 4210-6 standards [4]. The FEA tests conducted were as follows:

1. Horizontal loading scenario with a dynamic loading force of 600 N at a frequency of 25 applied on each end of the head tube [4].
2. Static start-up in a hill climb (Overweight rider) and axial loading forces of 143.6 N inwards and 1090 N downwards on each side of the bottom bracket [4].
3. Steady State scenario with axial loading forces of 20 kgf at each side pedal connected to the bottom bracket, 70 kgf at the end of the seat tube [4].

Further research was conducted to understand the relationship between the beam cross sections and the corresponding stress and displacements undergoing loading. Figures 16 to 20 show the

cross section options for each of bicycle frame components. All cross sections were positioned vertically and a fixed constraint was applied to the bottom end of the tube and the loads (Bending load of 250N, Torsional load of 50 Nm) was applied to the top end of the tube [4]. The cross sections were then compared based on displacement, mass, weighted drag, production/cost effectiveness as shown in Table 6.

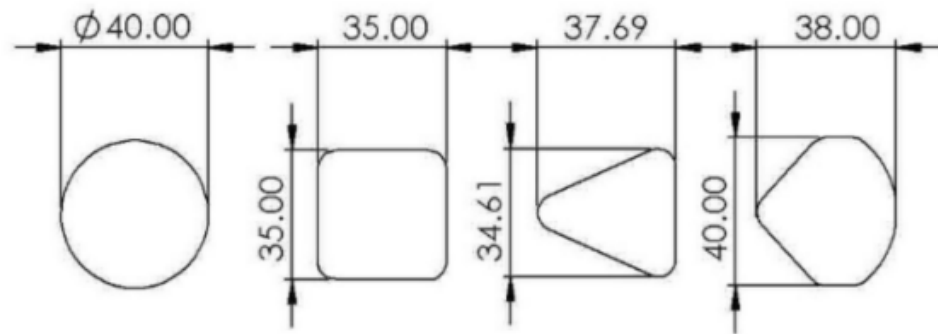


Figure 16: Top Tube Cross sections [4]

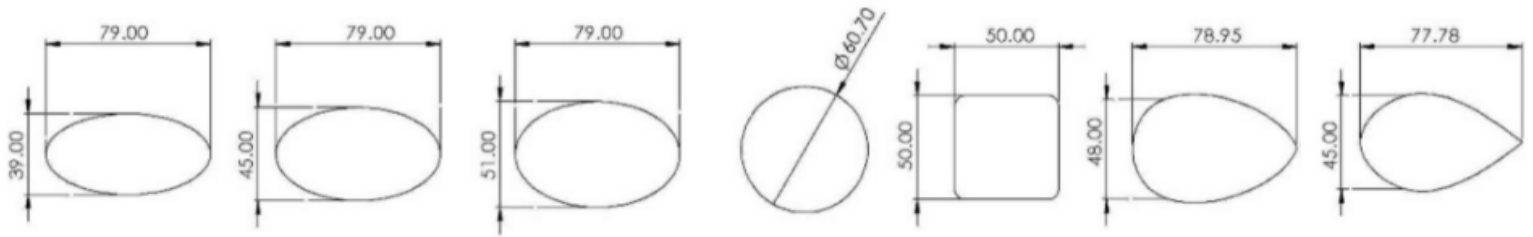


Figure 17: Down Tube Cross-Sections [4]

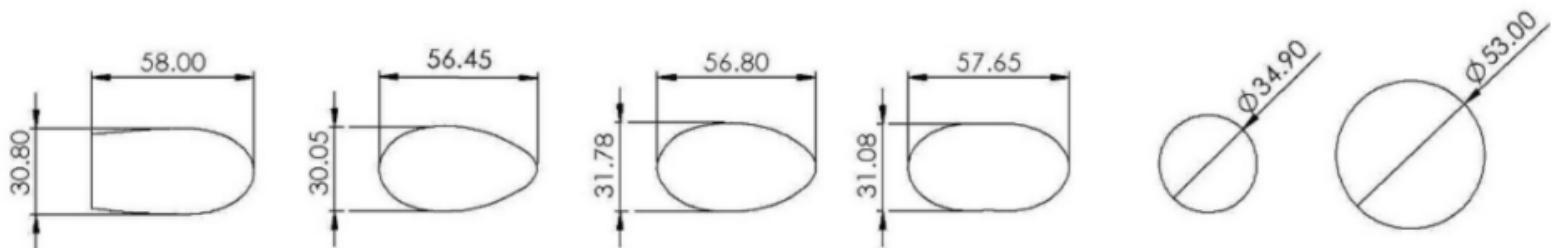


Figure 18: Seat Tube Cross sections [4]

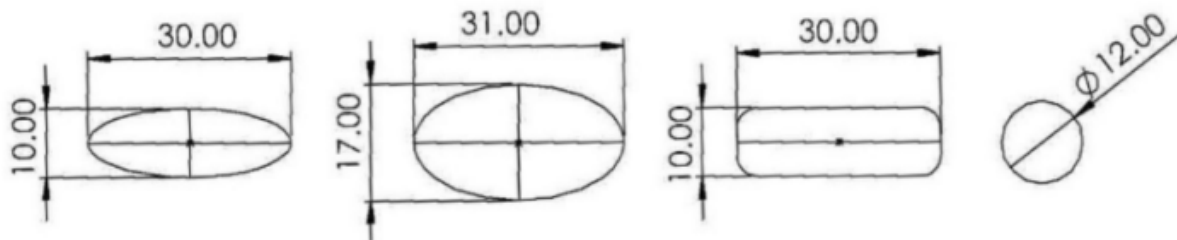


Figure 19: Seat Stat Cross Sections [4]

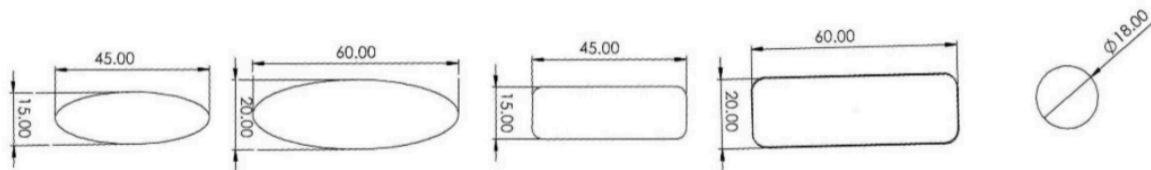


Figure 20: Chain Stay Cross Sections [4]

Aside from the bicycle frame, the bicycle transmission was also researched to understand how the bicycle is able to move with forces from the human foot. The transmission of the bicycle consists of several components and gears which transforms pedaling movement into movement of the bicycle through the rear wheel. There have been several types of transmission systems used in bicycles over the years. These types include: toothed belts, cardan shafts, however, the most common and widely used transmission system in bicycles is the chain based transmission system, as shown in Figure 21. The advantages of employing a chained transmission system will ensure that the bicycle avoids slippage, will transmit large forces, and will make sure the gear shifts to drive the chain on pinions and sheets [5]. However, some of the drawbacks include noisy transmission, high weight, relatively expensive to produce, and requires chain maintenance (lubrication) [5].



Figure 21: Chain transmission [8]

The toothed belt transmission, also known as the “carbon belt drive”, has replaced the ‘chained’ mechanism with bike models. Instead of the typical pinion and sheet, the two gears are mounted onto the shaft [8]. The transmission belt strap is composed of carbon fiber rubber. Compared with the chain transmission, the advantages of using the toothed belt transmission include low wear, minimal maintenance, silent running transmission, lower weight, and high durability [8]. Some of the setbacks are that it is weather sensitive, and is generally expensive due to carbon fiber reinforcement [8].



Figure 22: Toothed Belts Transmission [8]

Cardan (Chainless) shaft transmission was a traditional method of transmission. The movement of the bicycle is also transmitted to the rear wheel, but with the shaft instead of a ‘chained’ mechanism. Some of the advantages of using this transmission are that it provides a simplistic and aesthetic look for the racing bicycle, transmits high moments without slides, it is well protected from weathering conditions, and provides a silent running transmission. However, the drawbacks from using this transmission is the expense of buying the bicycle with the type of transmission since it is not commonly accessible, and is high in weight.



Figure 23: Cardan Shaft Transmission [8].

Analyzing the various types of tires used on bikes is a crucial aspect in determining bicycle performance. Different tires provide different kinds of driver performance outcomes depending on varying amounts of internal pressures, vertical forces, speed, and rolling surface temperature which affects the mechanical characteristics of a racing bicycle [8]. One of the characteristics the tires provide for the bike riders is performance optimization, where tire characteristics like tread pattern, width, inflation pressure are analyzed to elevate grip, minimize rolling resistances, and strengthen against punctures on the track. This examination not only fine-tunes the bicycle's performance, but also improves the bicycle's overall handling performance for the rider's experience [8]. Moreover, the tire analysis also relies on durability and reliability on the types of tires. Tire analysis explores opportunities to trim weight without sacrificing durability while also enforcing rolling resistance to enhance bicycle's efficiency and rider's comfort [8]. Furthermore, the nature of the empirical examination from tire analysis leads to tire customization, allowing

bicycle manufacturers to tailor tires to the needs of certain bicycle types and riding preferences [8]. Thus, whether it's for road racing, mountain biking, commuting, or touring, tire analysis ensures that each bicycle is equipped with the optimal tires to elevate its performance, safety, and rider satisfaction [8].

Some of the common bicycle tires widths that are current sold in the markets are as follows:

1. Mountain Bike Tire Width
2. Gravel Bike Tire Width
3. Road Bike Tire Width
4. Commuter Bike Tire Width



Figure 24: Mountain Bike Tire Width [9].

Figure 24 depicts the mountain bike tire width. The tires of mountain bikes feature assertive tread designs tailored to enhance traction on challenging surfaces like loose dirt, muddy terrain, and rocky trails. Various mountain biking disciplines require distinct tread patterns optimized to excel in specific environmental conditions.



Figure 25: Gravel Bike Tire Width [9].

Striking a balance between road and mountain tire designs, gravel tires boast a tread pattern that combines speed with traction, such as illustrated in Figure 25. Numerous gravel tires sport a dense arrangement of small knobs spanning the tire's surface, while others opt for a smooth center tread complemented by grippier sections along the edges, facilitating confident cornering on loose gravel surfaces.



Figure 26: Road Bike Tire Widths [9].

Figure 26 depicts the road bike tire width pattern. Slick tires are characterized by an almost imperceptible tread pattern, boasting a smoothness that is barely discernible. Positioned at the center of the tire, slick rubber is engineered for swift rolling on paved surfaces, while subtle tread along the tire's edges serves to disperse water and enhance traction during cornering maneuvers.



Figure 27: Commuter Bike Tire Width [9].

Featuring a semi-slick tread design, the commuter bike tire width, such as depicted in Figure 27, excels across various terrains and weather conditions. The smooth center tread ensures swift rolling on city streets or paved bike paths, while the slightly more textured sides, equipped with small knobs or deep channels, enhance grip and water dispersion. This configuration enables commuters to confidently navigate wet conditions or tackle occasional off-road stretches with ease and safety.

1.5. Objectives and Design Requirements

The objective of the design requirements section is to leverage the use of computational analysis toolkits as a way for engineers to optimize the design of a prototype by predicting new design configurations, performance outcomes, and fine-tune parameters with precision and efficiency. Taking the approach for utilizing computational analysis for designing a new bicycle frame

requires in depth analysis and testing/validation when using computer aided design parameters. The systemic procedures are illustrated in Figure 28 below as a basic guideline for optimizing the design of the racing bike using the software toolkits based on marketing trends towards production of the prototype.

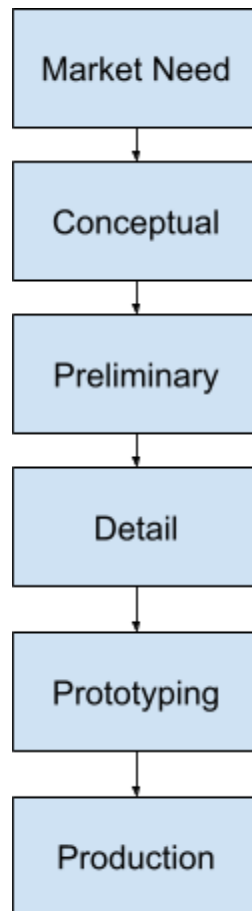


Figure 28: Design Requirements Flowchart.

2. PROBLEM DEFINITION

2.1. Statement of the Problem

In the world of traditional bicycles, engineers face challenges that necessitates a focus on innovation and improvement. Despite its evolution due to increase in modern technology, conventional bicycles continue to grow with various limitations and inefficiencies that lack their overall performance and usability. Some of the challenges include factors such as suboptimal aerodynamics, weight distribution, structural integrity, and rider comfort. Additionally, with the growing demand for sustainable transportation solutions and the increasing emphasis on efficiency and eco-friendliness, there is a growing trend for sustainable solutions and markets are on the lookout to cater to their business needs. Therefore, engineers must prioritize innovation in this domain to address these challenges effectively, leveraging advancements in materials science, aerodynamics, ergonomics, and computational modeling to develop bicycles that are not only faster, lighter, and more comfortable but also more environmentally sustainable.

2.2. Requirements Analysis

Table 7: Requirements chart based on SUC.

SUC	Requirements
Setting Up/Initialization	• Bike must be used in all-terrain (sloppy, grassy, sandy, etc.)
Using the Intervention	• Bike must be able to travel up and down slopes, thus the vehicle must have mechanisms to prevent rolling down uncontrollably.
Finalization	• Bike must be able to balance and maneuver in all-terrains. • Bike must fit in narrow spaces. • Bike must travel as smoothly as possible in all-terrains.

Table 8: Requirements Validation Chart based on SUC.

SUC	Type	Requirements
External Validation	Usability/Mobility	• Bike must have places to store items such as a laptop. • Bike must be used in all-terrain (sloppy, grassy, sandy, etc.)
	Maneuverability/Turn	• Max Turning radius: 0ft. • A tight turning radius is important in order to effectively navigate difficult terrain.
	Height of Seat/Bike	• Must be able to have a height of at least 1350mm (This indicates that a height of 1350 mm will put the average user under the standing eye level of the average person).
	Wheel Size	• The wheel size must be able to turn tight spaces and size affects the turning radius
Internal Validation	Comfort	• Bike must accommodate users' usage in all-terrain. • Bike must be structurally designed to make it comfortable for the users and simultaneously durable with usage.
	Control	• Bike must be controllable by the users in terms of speed and total control. • Bike must be adjustable according to the user's preferences and certain situations
Constraints	Weight	• Bike must be able to withstand an external load of at least 500 lb.

2.3. Evaluation of Existing Designs

It is crucial to evaluate existing racing bicycle frames to identify strengths, weaknesses, and opportunities for improvement. This involves a comprehensive analysis of frame geometry, material composition, and performance characteristics. Existing designs are scrutinized through a combination of literature review, field observations, and benchmarking against industry standards. Key performance metrics such as weight, stiffness, aerodynamics, and rider feedback are considered to gauge the effectiveness of each design in meeting the demands of competitive cycling. Through this evaluation process, insights are gained into the prevailing trends and design philosophies in the racing bicycle industry. Additionally, it provides a foundation for establishing design objectives and criteria for the development of the new racing bicycle frame.

Furthermore, taking into consideration the past bike designs, the integration of computer-aided design softwares has emerged as a transformative solution to the engineering challenges as a method to save such investments as previously discussed in the past approaches. By leveraging the use of computational analysis toolkits, engineers are able to optimize the design of a prototype by predicting new design configurations, performance outcomes, and fine-tune parameters with precision and efficiency. Taking the approach for utilizing computational analysis for designing a new bicycle frame requires in depth analysis and testing/validation when using computer aided design parameters. The systemic procedures are illustrated in Figure 29 below as a basic guideline for optimizing the design of the racing bike using the software toolkits.

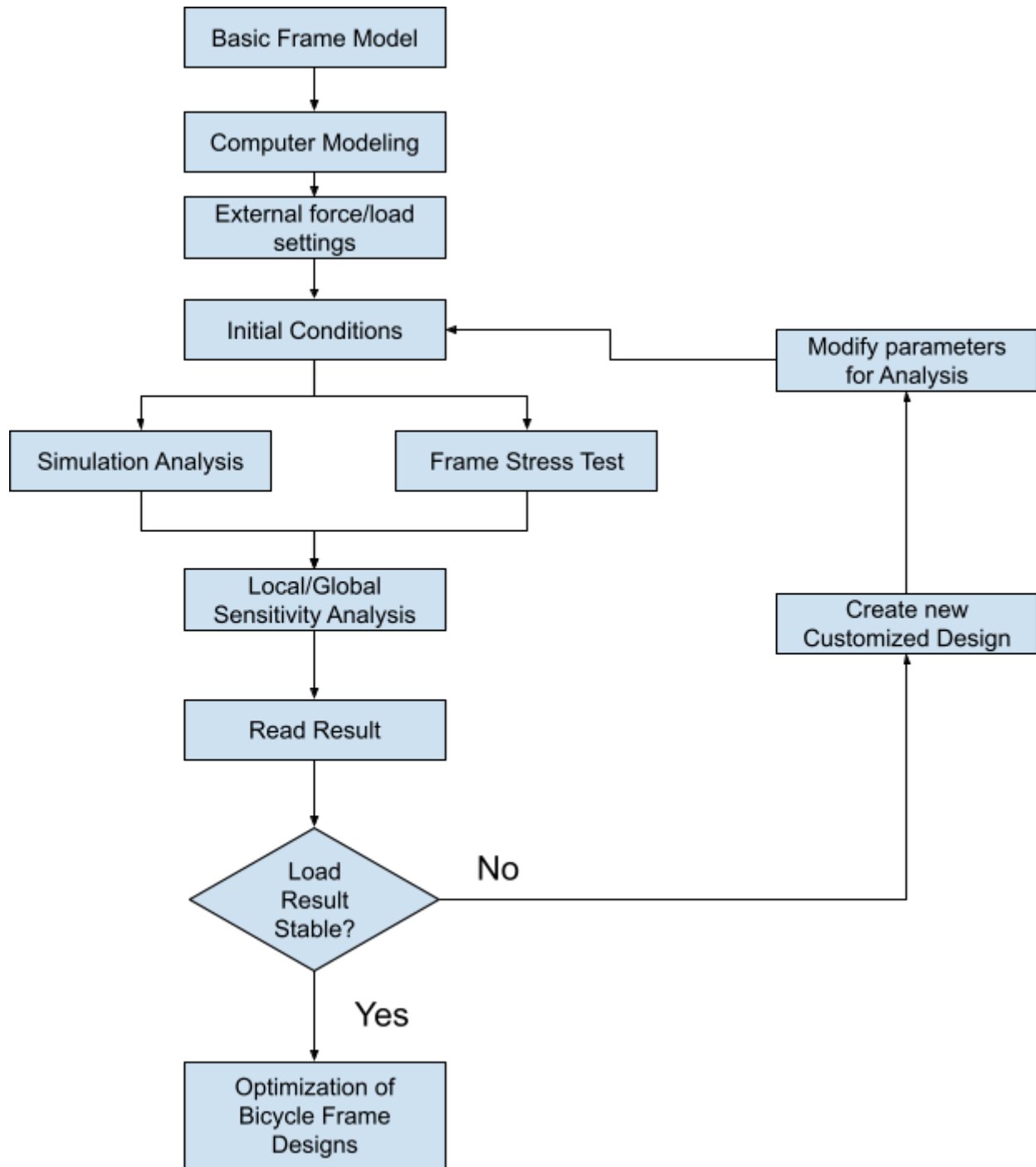


Figure 29: Flowchart of the Iterative Design Procedure for using Computer Aided Design into modeling the Final Design of the Prototype.

2.4. UCI Regulations of Racing Bicycles



Figure 30: International Cycling Union (UCI) [2].

The UCI serves as the overseer responsible for ensuring the appropriate adherence to ethical and sporting rules [1]. Moreover, the UCI Regulations emphasize the supremacy of human effort over mechanical advantages. Adherence to these regulations by all participants ensures equitable competition and enhances safety in all aspects of the rider's experience.

Some of the regulations by UCI are listed below:

- “Each license holder shall ensure that his equipment (bicycle with accessories and other devices fitted, headgear, clothing, etc.) does not, by virtue of its quality, materials or design, constitute any danger to himself or to others.”
- “The bicycle is a vehicle with two wheels of equal diameter. The front wheel shall be steerable; the rear wheel shall be driven through a system comprising pedals and a chain. Exceptions to this rule may exist for certain cycling disciplines, in which case specific rules are provided for in the respective discipline.”
- “The bicycle should have handlebars which allow it to be ridden and maneuvered in any circumstances and in complete safety.”
- “A bicycle shall not measure more than 185 cm in length and 50 cm in width overall. A tandem shall not measure more than 270 cm in length and 50 cm in width overall.”
- “The peak tip of the saddle shall be a minimum of 50 mm to the rear of a vertical plane passing through the bottom bracket spindle.”
- “The distance between the bottom bracket spindle and the ground shall be between 24 cm minimum and maximum 30 cm.” - The rule is stated in order to avoid the risk of the cranks or pedals touching the ground when cornering.

- “The distance between the internal extremities of the front forks shall not exceed 11.5 cm; the distance between the internal extremities of the rear triangle shall not exceed 14.5 cm.”
- “Wheels of the bicycle may vary in diameter between 70 cm maximum and 55 cm minimum, including the tyre. For the cyclo-cross bicycle the width of the tyre (measured between the widest parts) shall not exceed 33 mm and it may not incorporate any form of spike or stud.”
- “The weight of the bicycle cannot be less than 6.8 kilograms.”
- “For road, track, and for cyclo-cross competitions, the frame of the bicycle shall be of a traditional pattern, i.e. built around a main triangle.” Such as depicted in Figure 4.

2.5. Competition Event in Mountain Biking

Different types of racing conditions demand varying engineering considerations and optimization strategies to ensure performance across terrains and different types of racing conditions. Cross-country (XC) events encompass a range of disciplines, each presenting unique challenges for engineers and riders alike. As such, Cross-country Olympic (XCO) races typically feature technical singletrack trails and steep climbs, necessitating lightweight frames, efficient power transfer, and agile handling characteristics to navigate tight corners with precision. Cross-country marathon (XCM) events, on the other hand, emphasize endurance over long distances, requiring designs that prioritize comfort, durability, and energy efficiency to withstand extended periods of riding. Point-to-point races (XCP) present a blend of technical descents and long climbs, necessitating versatile designs capable of maintaining speed and stability across diverse terrain. Short track events (XCC) demand quick acceleration and sharp handling to navigate tight circuits and frequent obstacles, calling for agile and responsive frames optimized for agility and maneuverability. Eliminator races (XCE) introduce head-to-head competition on short, technical courses, highlighting the importance of acceleration, agility, and robust construction to withstand the rigors of intense competition. Time trial races (XCT) prioritize aerodynamics and efficiency, requiring designs that minimize drag and maximize power transfer to achieve maximum speed over timed segments. Finally, team relay events (XCR) demand reliable and adaptable designs capable of accommodating multiple riders and quick transitions, emphasizing durability, versatility, and ease of maintenance. In all racing scenarios, engineering considerations such as frame geometry, material selection, suspension design, and component integration play a crucial role in optimizing performance, enhancing rider experience, and achieving success in competitive mountain biking. Table 8 presents an overview of the racing types each with their racing times and distances.

Table 9: Types of cross-country racing biking events (race time and/or distance are not well defined or described by UCI regulations.)

Event	Race Time (min)	Circuit Distance (km)
XCO	80-100	4-6
XCM	60-100	20-160
XCP	20-30	
XCC	20-40	~2
XCE	2-5	1
XCT	30-50	4-6
XCR	200-720	-
XCS	-	-

3. CONCEPT DESIGN AND EVALUATION

3.1. Design Strategy

Iteration 1: 7 - Member Bicycle Frame Design

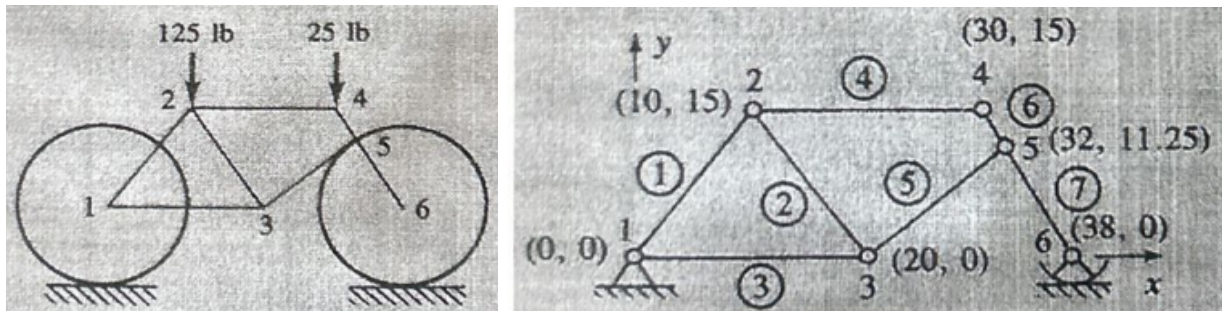


Figure 31: Typical Bicycle frame geometry prepared by Dr. J.Cao

In iteration 1, a simple loading condition of 125 lb and 25 lb was applied to the end points of the top tube as shown in Figure 32. The following analysis in Figures 32, and 33 describes the deflection of the truss bicycle under loading conditions and corresponding nodal solution of the truss displacement under load respectively. From the nodal solution of the 7-member truss bicycle, it can be observed that the fork tube and head tube are the main areas of concern as shown in Figure 31.

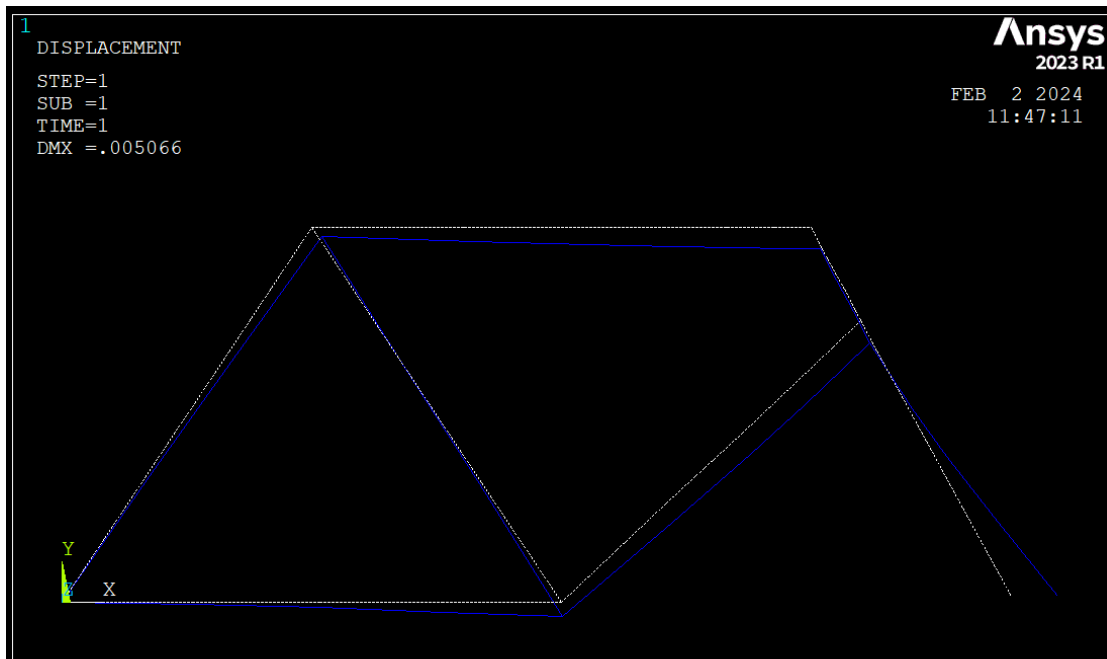


Figure 32: Deflection of a typical 7-member (2 Dimensional) truss bicycle under load.

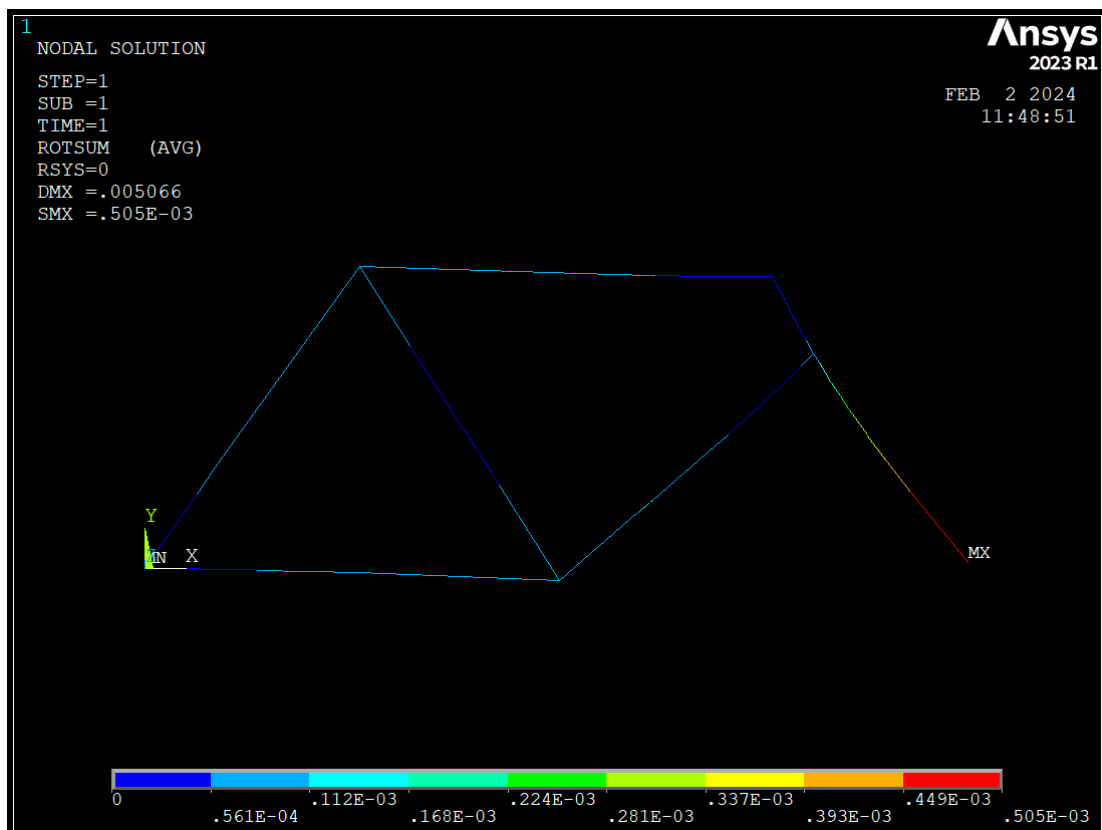


Figure 33: Nodal solution of a 7-member (2 Dimensional) truss bicycle under load.

Iteration 2: 7 - Member Bicycle Frame Design with the 5 Loading Conditions

In iteration 2, the identical 7-member bicycle frame was used to simulate the behavior of the bicycle frame under the 5 loading condition approach according to Sarath P, et al. and Akhyar, et al [5, 6]. The five loading conditions are Static Start-up, Steady State Pedaling, Vertical Impact, Horizontal impact and Rear Wheel Braking and the corresponding loading condition values and locations are listed in Table 2 under the *Literature Review Section*. In the ANSYS simulation for each corresponding loading condition, 3 solutions will be generated: Elements, Displacement and Nodal Solution.

Loading Condition 1: Static Start-Up

A load of 700 N and 200 N applied to the top of the seat tube and bottom of the bracket respectively.

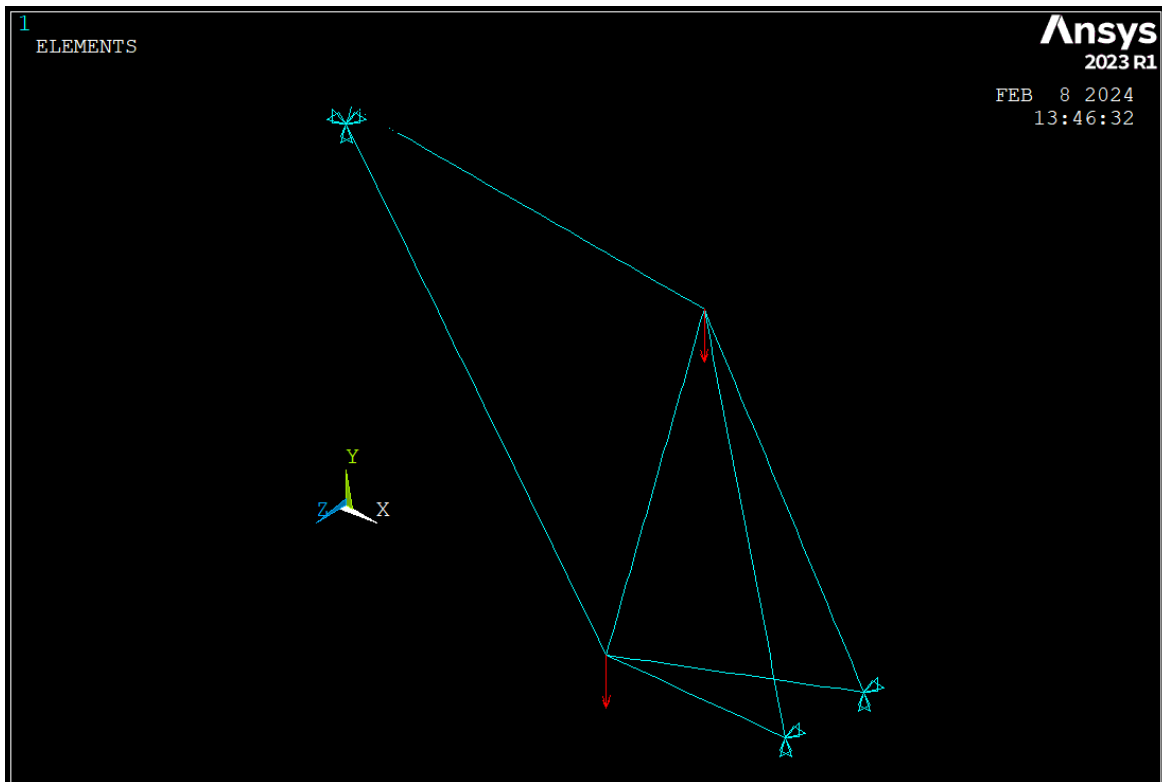


Figure 34: 7-Member truss bicycle frame under Static Start-up loading

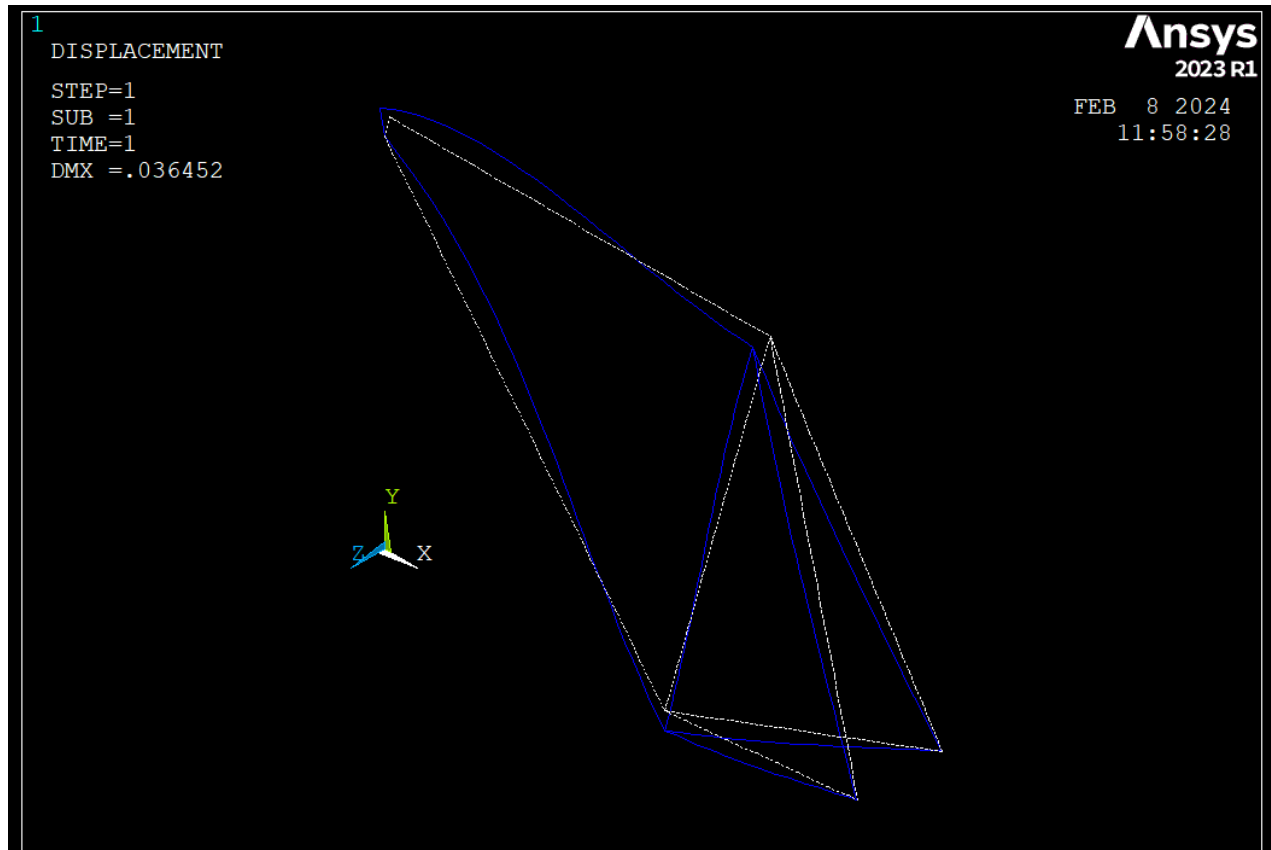


Figure 35: Deflection of 7-Member truss bicycle frame under Static Start-up loading

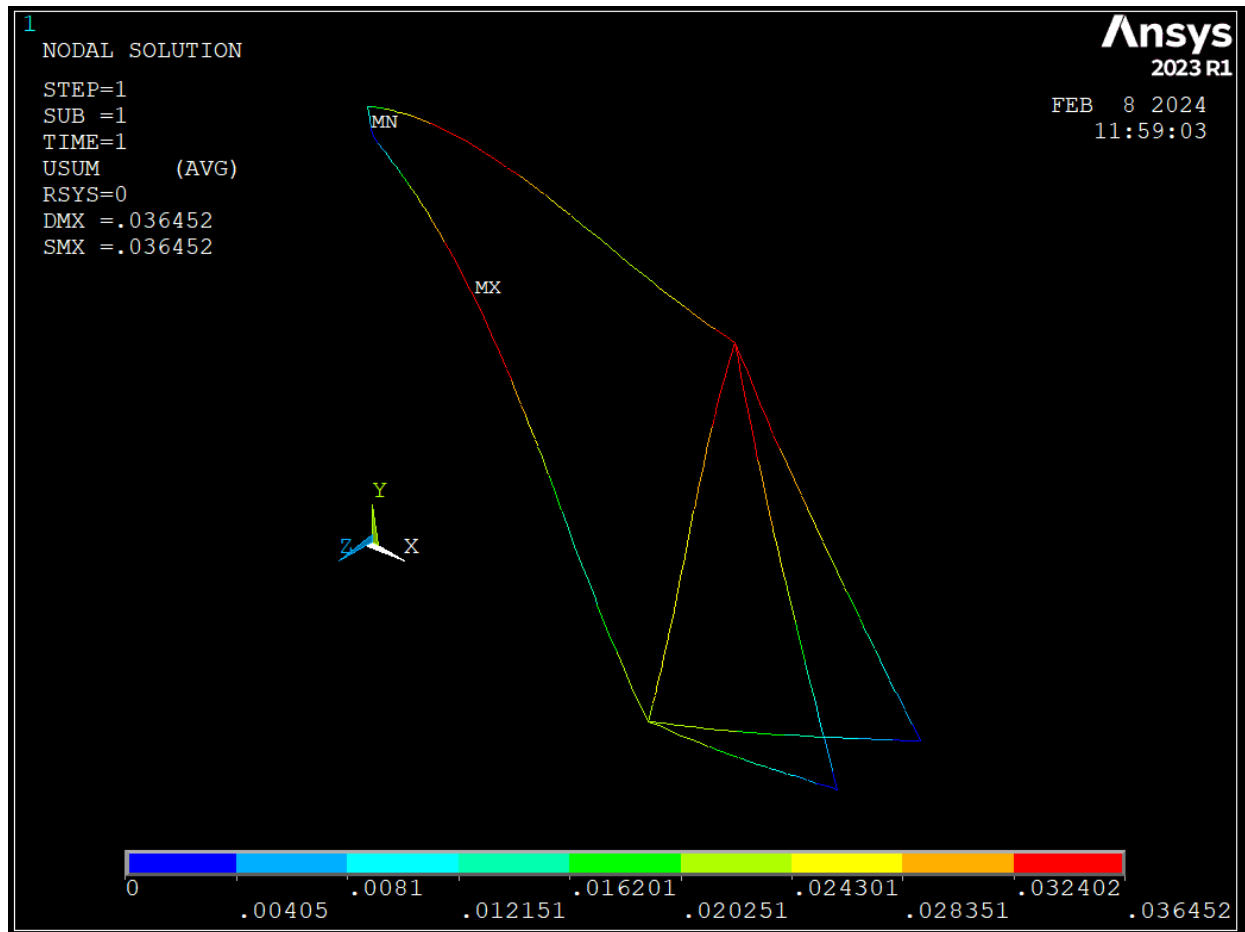


Figure 36: Nodal Solution of 7-Member truss bicycle frame under Static Start-up loading

From the nodal solution, it can be observed that the critical locations where maximum deformation is high are the top tube, down tube and the intersection of the seat stay, seat tube and top tube as shown in Figure 35.

Loading Condition 2: Steady State Pedaling

A load of 1000N and 200 N applied to the bottom tube and head tube respectively.

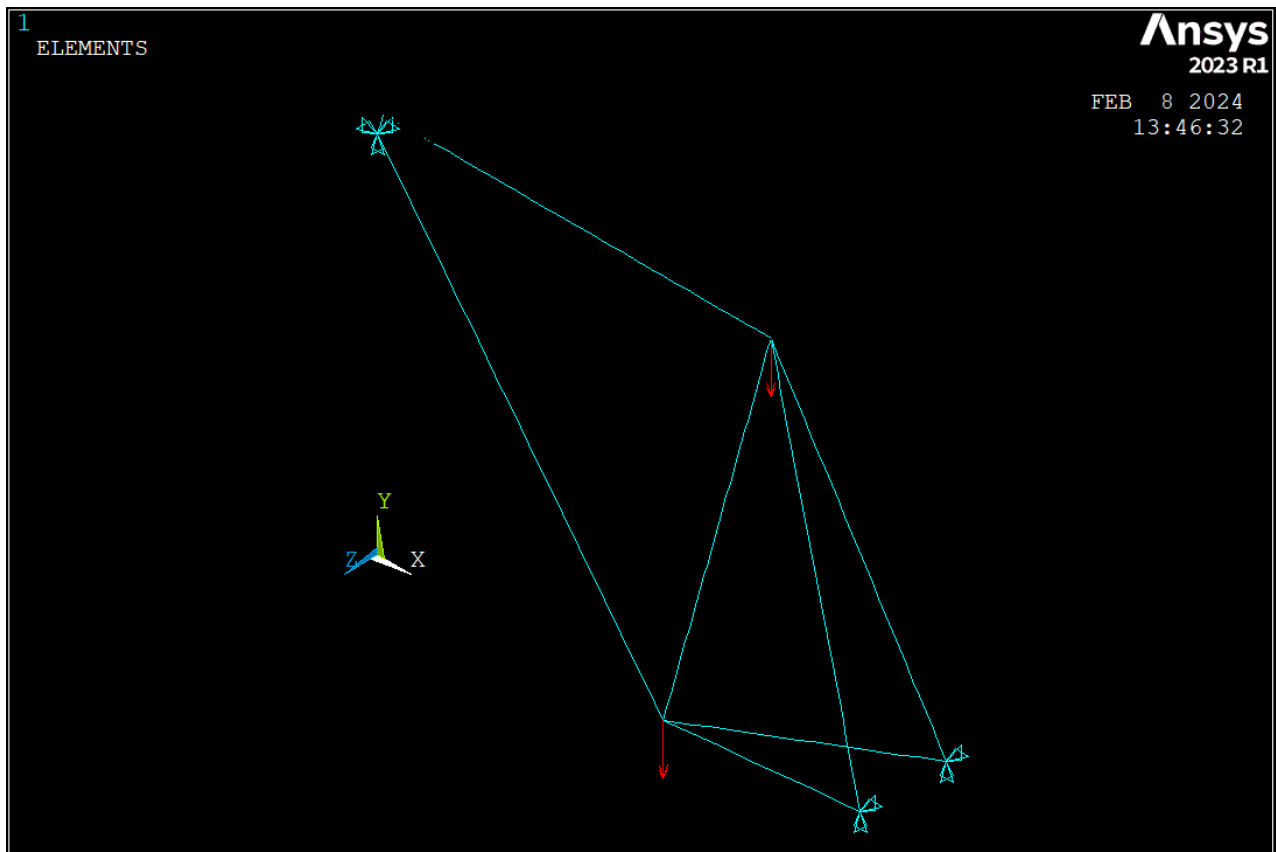


Figure 37: 7-Member truss bicycle frame under Steady State Pedaling loading

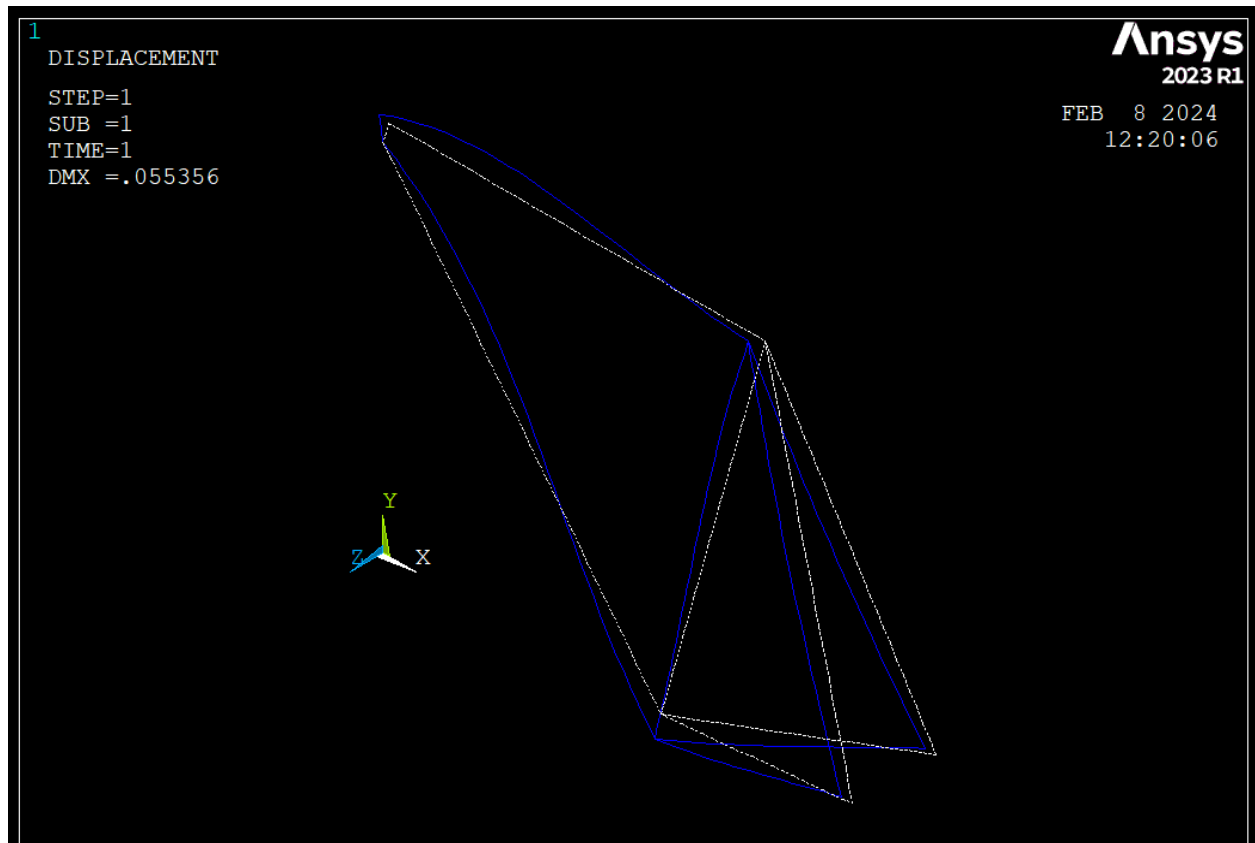


Figure 38: Deflection of 7-Member truss bicycle frame under Steady State Pedaling loading

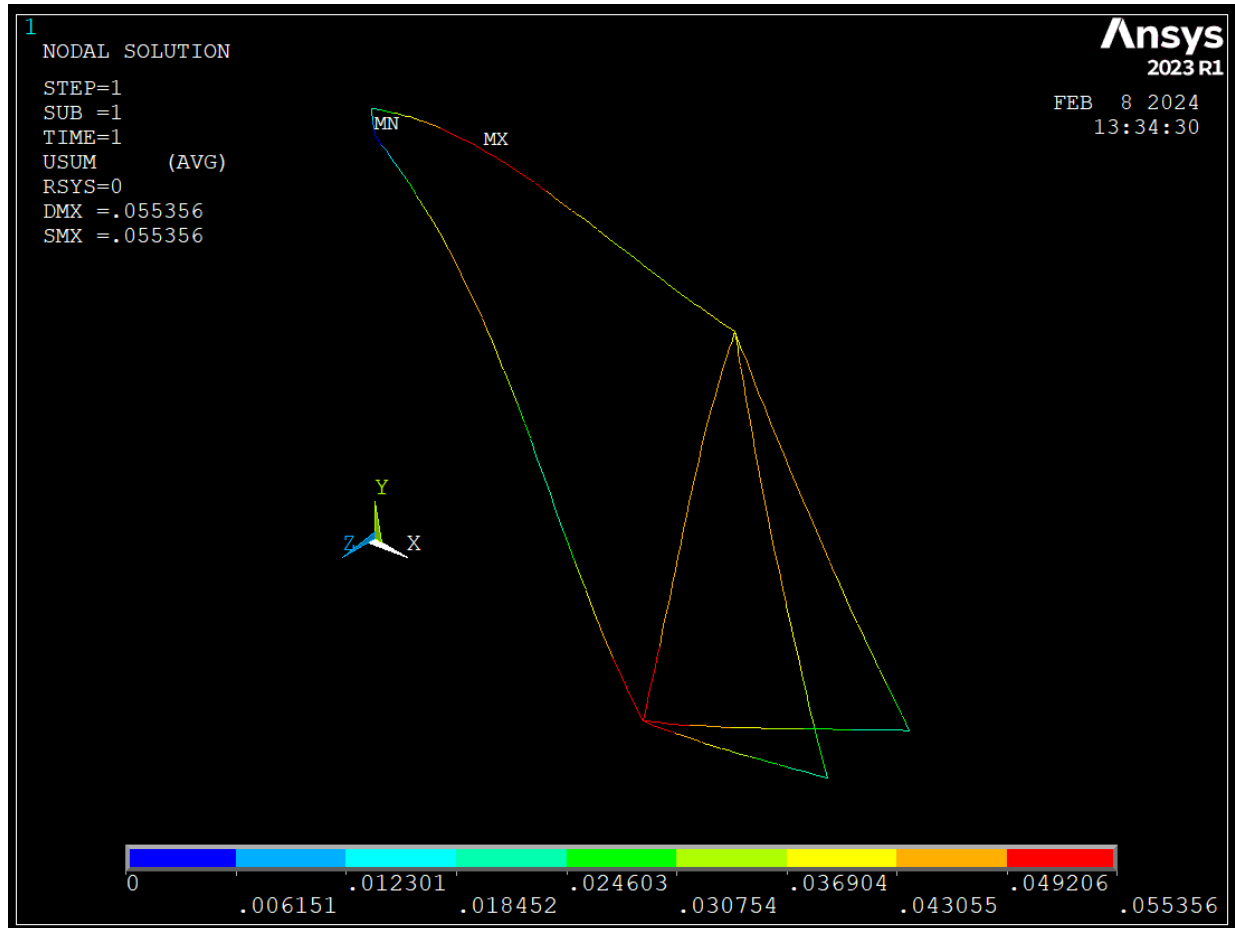


Figure 39: Nodal Analysis of 7-member truss bicycle frame under Steady State Pedaling loading

From the nodal solution, it can be observed that the critical locations where maximum deformation is high are the top tube, down tube and the intersection of the seat stay, seat tube and top tube as shown in Figure 39. This is our initial design model for a traditional bicycle model with the diamond frames intersecting at the seat tube (top point).

Loading Condition 3: Vertical Impact

A load of 2250 N applied to the top of the bracket. This is illustrated by applying a G factor to the cyclist's weight. Here, a factor of 2G is utilized, resulting in a load of 2250N, which is essential for scenarios where an object falls from a minimal height onto a solid surface.

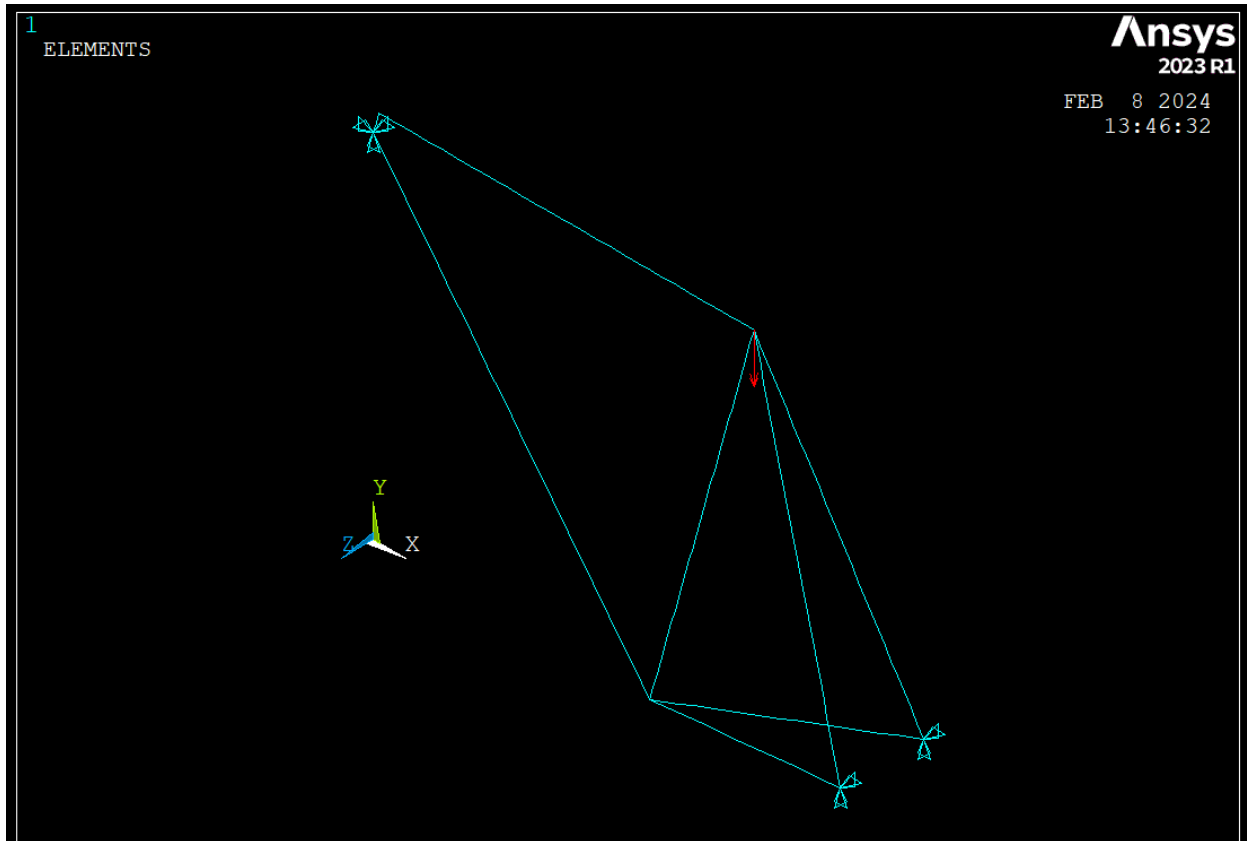


Figure 40: 7-Member truss bicycle under Vertical Impact Loading

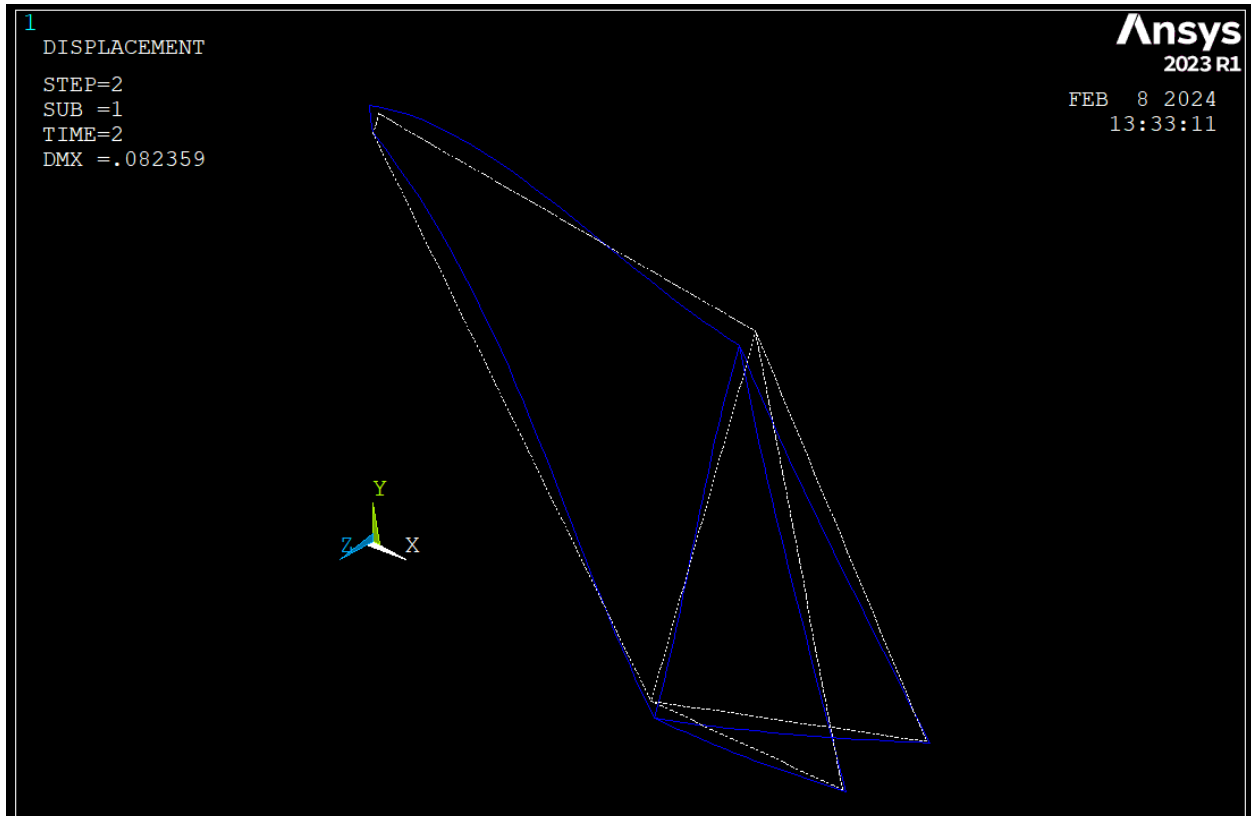


Figure 41: Deflection of 7-Member truss bicycle frame under Vertical Impact Loading

Vertical impact loading focuses on radial forces acting on the bicycle frame. The frames undergo sudden vertical forces, testing their ability to absorb shocks and maintain stability to avoid buckling or compromising rider safety as shown in Figure 41.

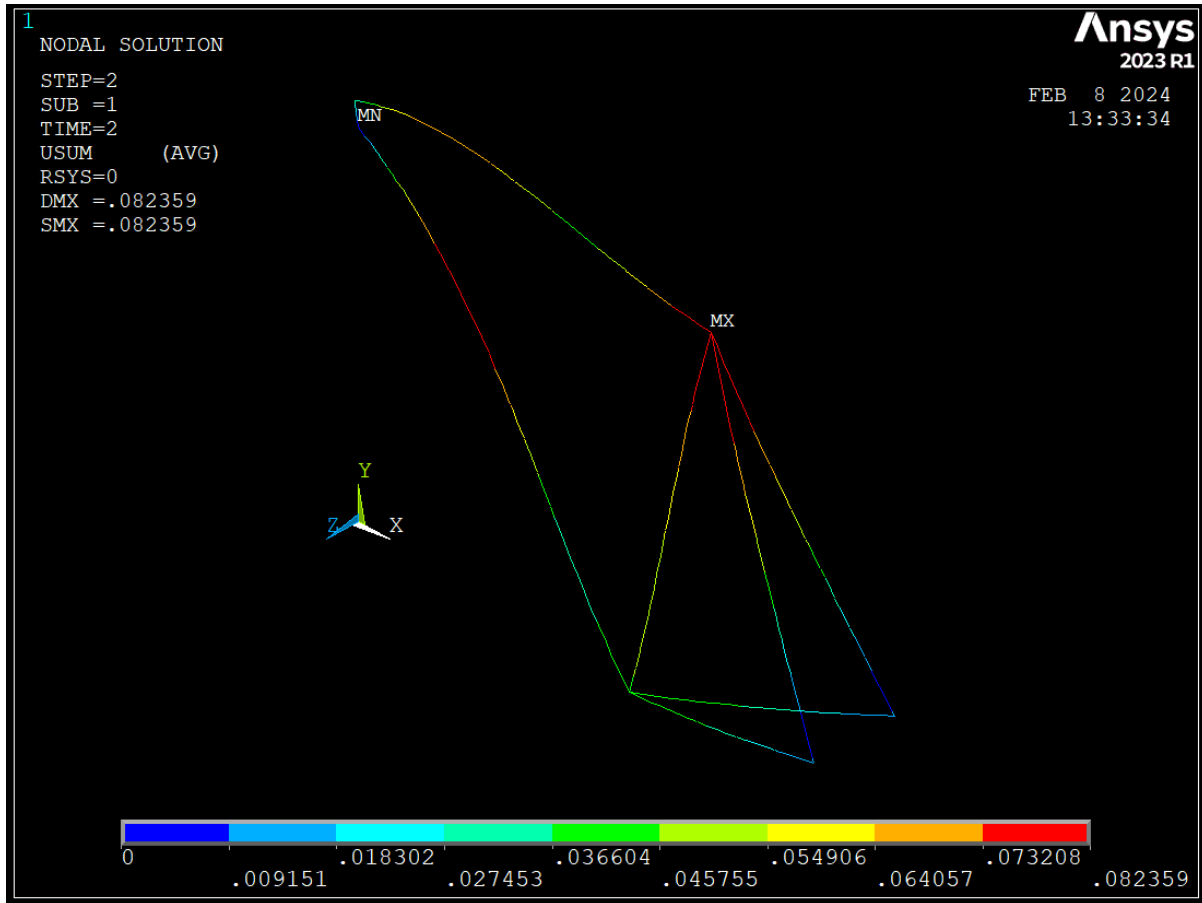


Figure 42: Nodal Analysis of 7-Member truss bicycle frame under Vertical Impact Loading.

From the nodal solution, it can be observed that the critical locations where maximum deformation is high are the down tube, the intersection of the seat stays, seat tube and top tube as shown in Figure 42. This is our initial design model for a traditional bicycle model with the diamond frames intersecting at the seat tube (top point).

Loading Condition 4: Horizontal Impact

A load of 2250 N applied to the head tube horizontally, the rear dropout is restricted from any translational movement.

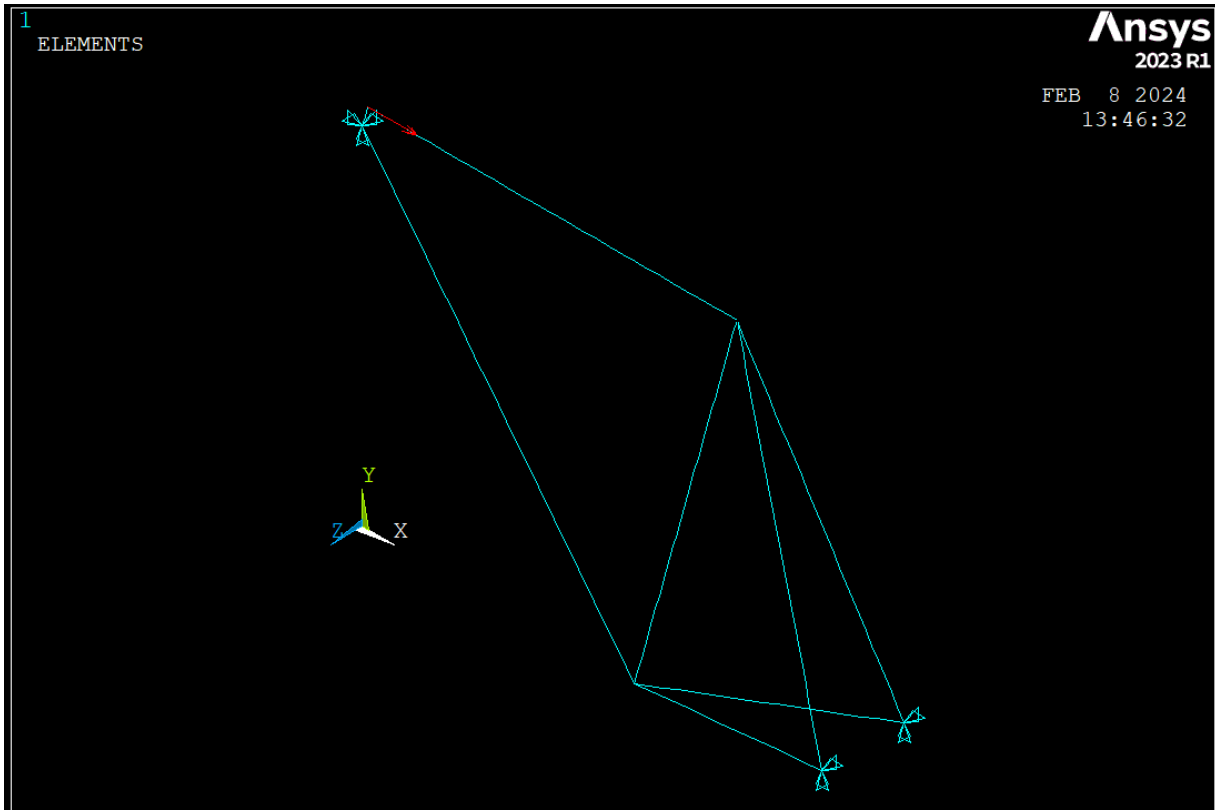


Figure 43: 7-Member truss bicycle frame under Horizontal Impact Loading

This loading scenario evaluates the frame's response to lateral forces, assessing its ability to withstand impacts.

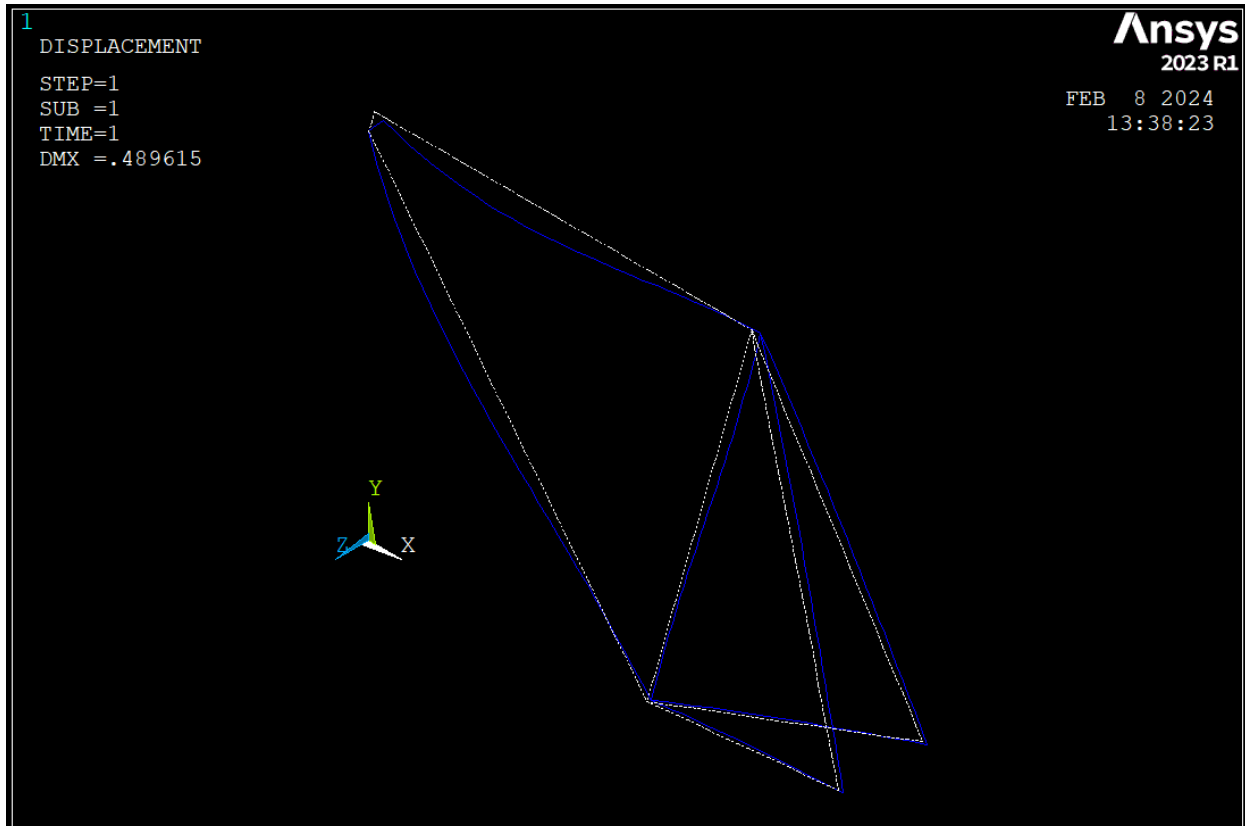


Figure 44: Deflection of 7-Member truss Bicycle frame under Horizontal Impact Loading

As illustrated in Figure 44, the Deformation Analysis determines if the frame structure can maintain rigidity and structural integrity under lateral loading.

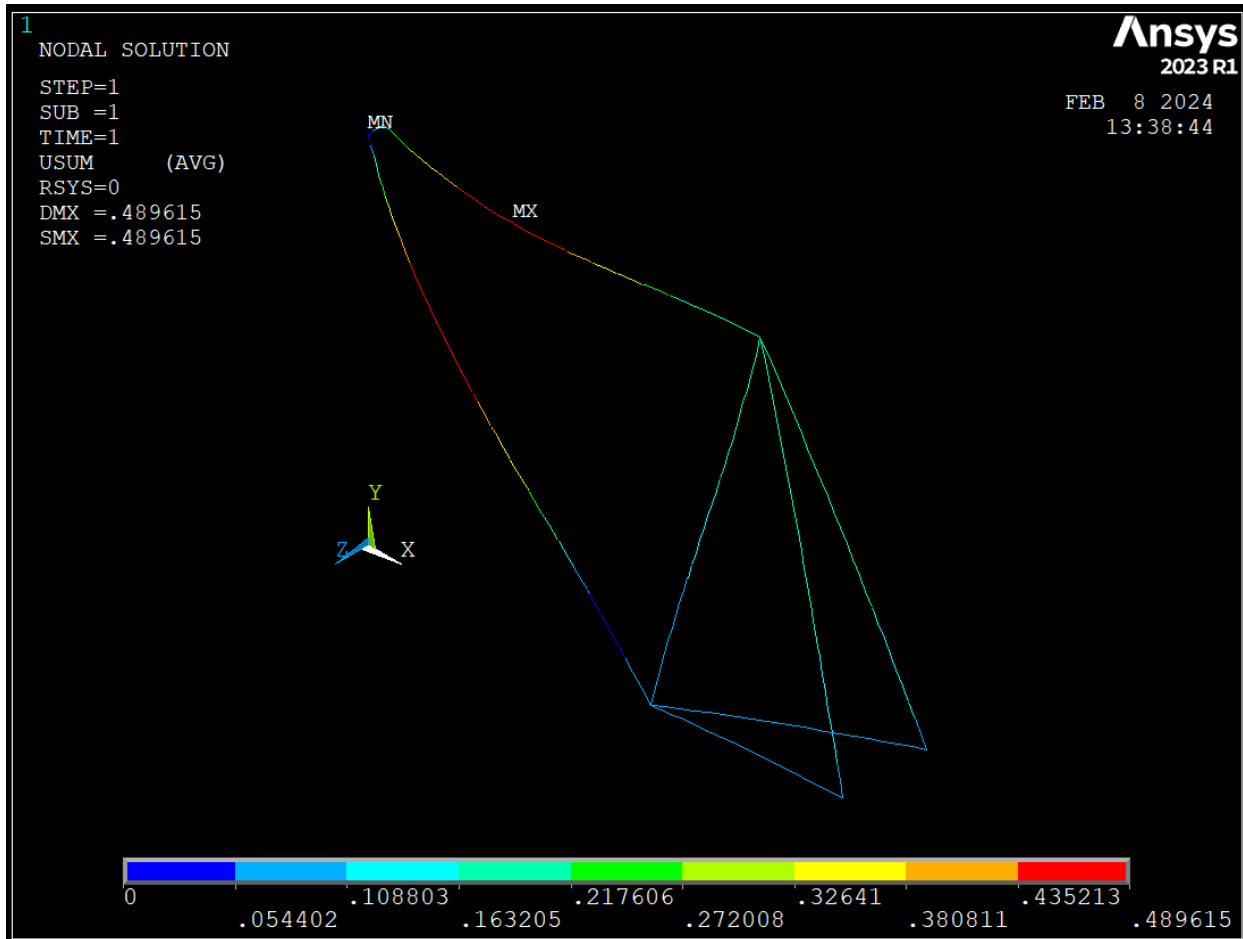


Figure 45: Nodal Analysis of 7-Member truss bicycle frame under Horizontal Impact Loading

From the nodal solution, it can be observed that the critical locations where maximum deformation is high are the top tube, and down tube as shown in Figure 45.

Loading Condition 5: Rear Wheel Braking

A load of 750 N applied to the rear wheel bracket. This loading condition simulates tire skidding, leading to an impending rear pitch-over.

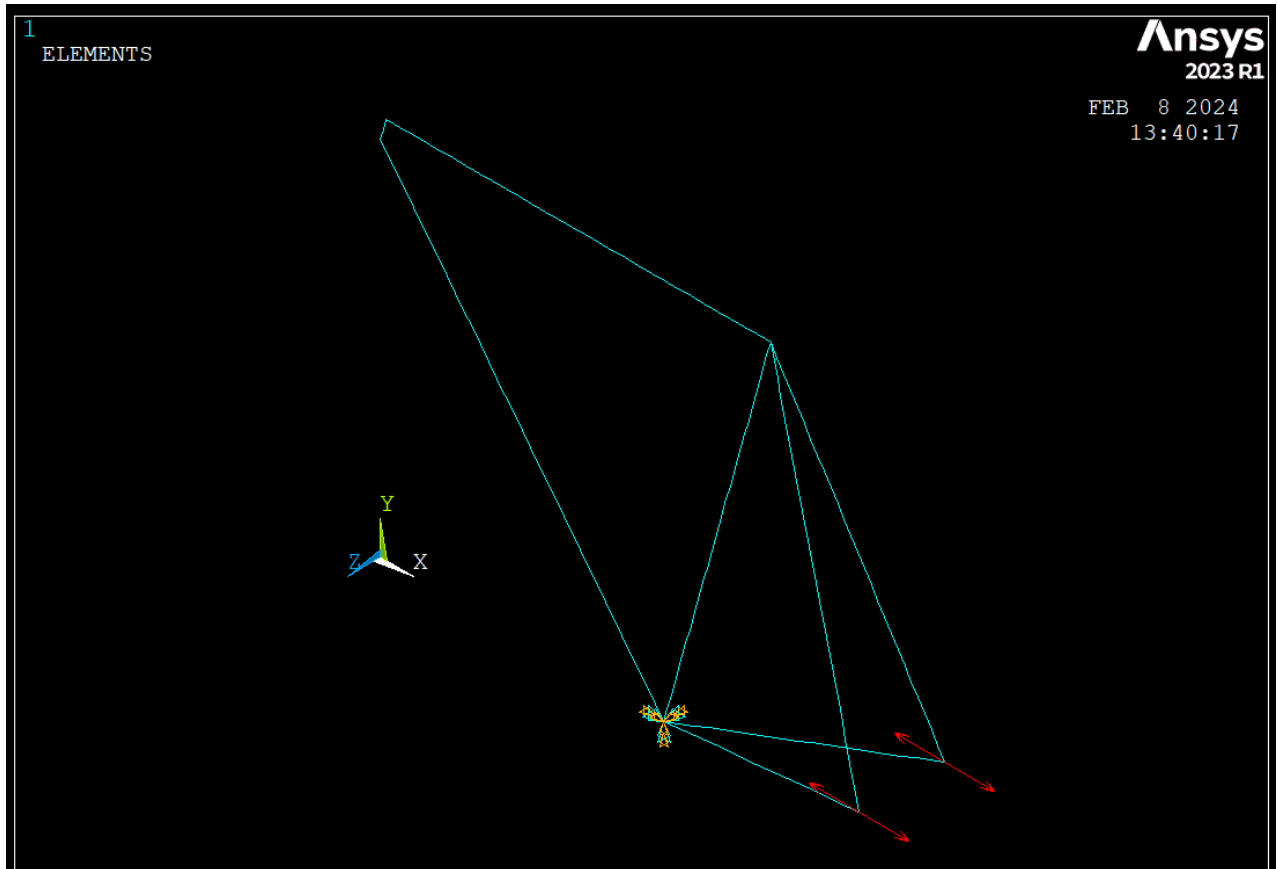


Figure 46: 7-Member truss bicycle frame under Rear Wheel Braking Loading

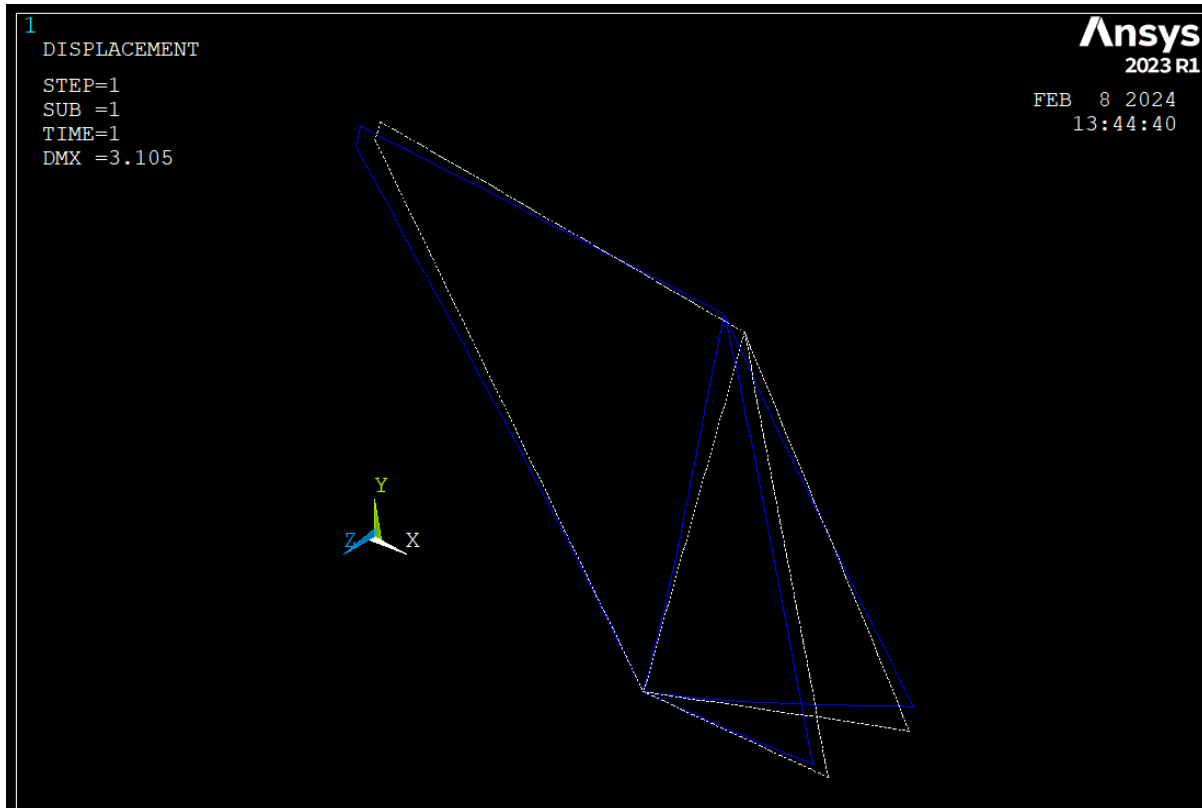


Figure 47: Deflection of 7-Member truss bicycle frame under Rear Wheel Braking Loading

From Figure 47, the frame experiences a combination of torsional and longitudinal forces as the rider applies the brakes at the rear end as shown in Figure 48.

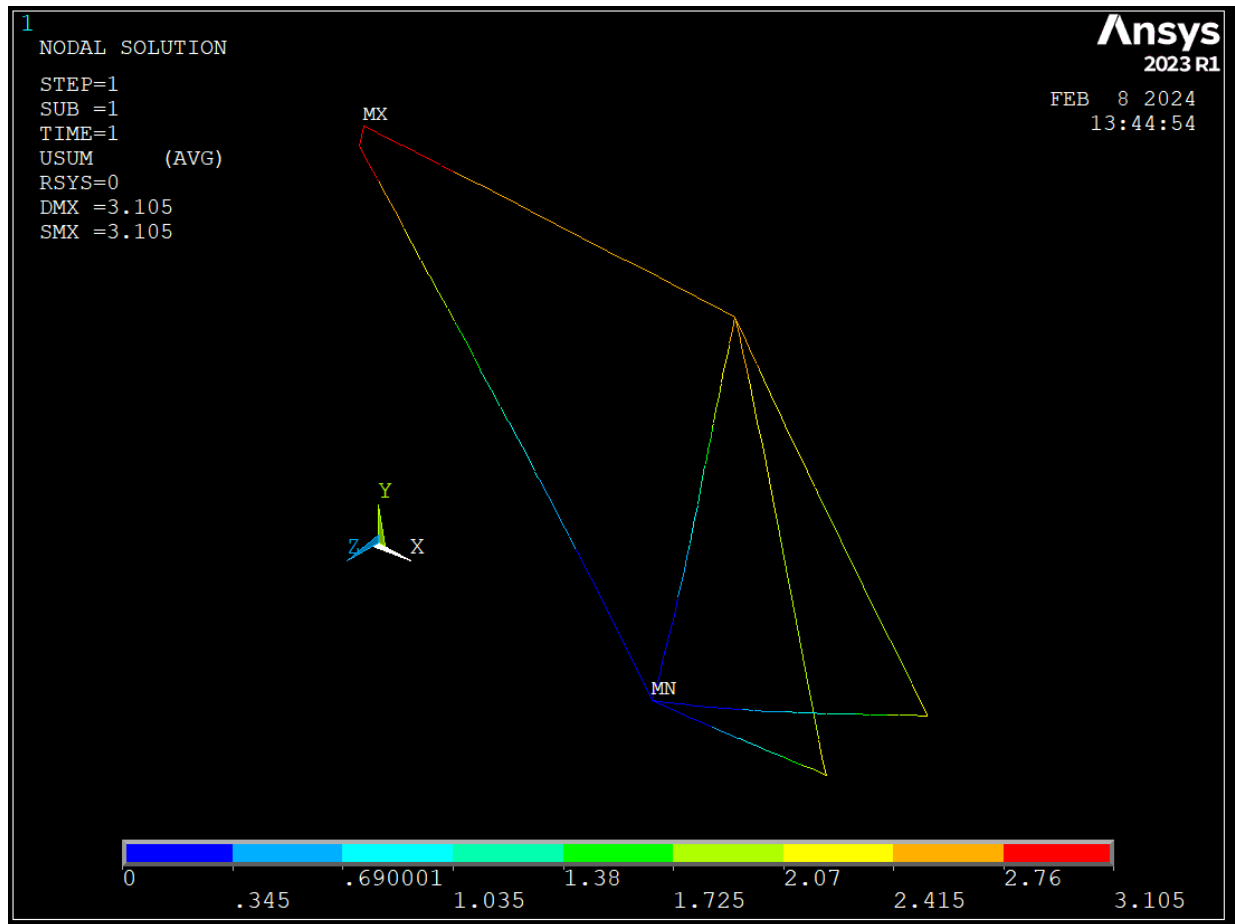


Figure 48: Nodal Analysis of 7-Member truss bicycle frame under Rear Wheel Braking Loading

From the nodal solution, it can be observed that the critical location where maximum deformation is high is the head tube.

The second iteration of the model is simulated using ANSYS Workbench with the similar loading conditions from Table 2 and is illustrated in Figures 49 and 54 below.

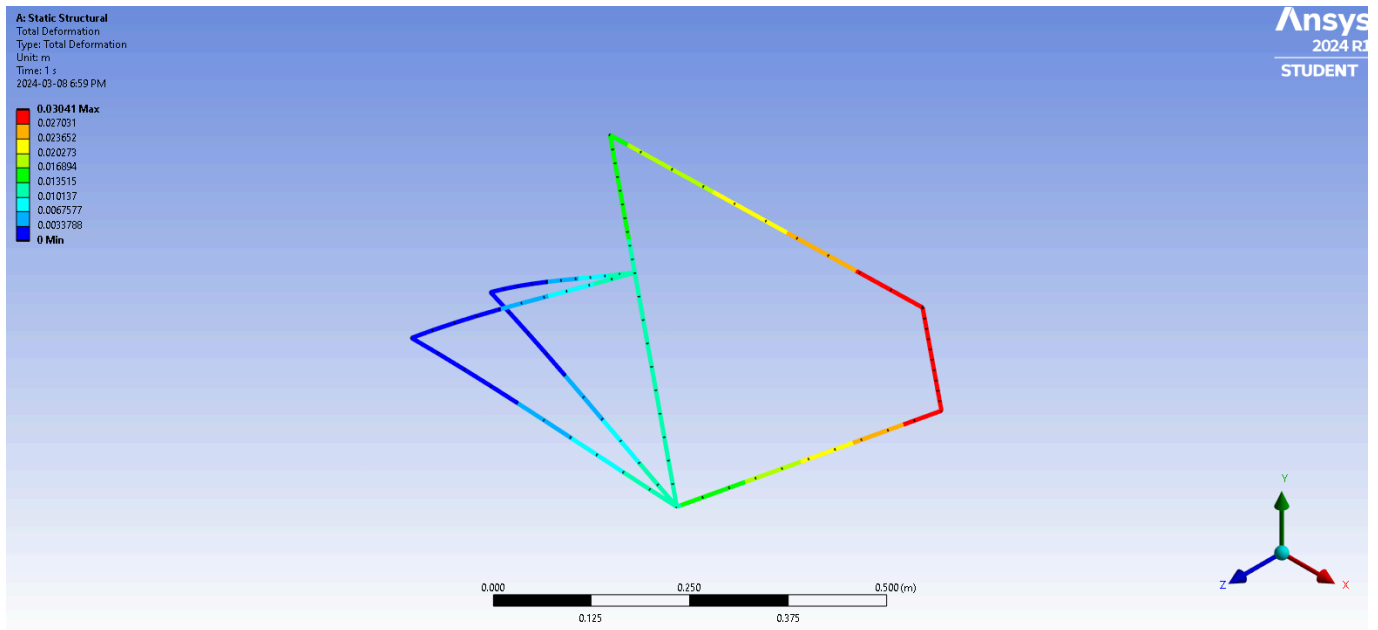


Figure 49: Max Deflection with wire frame in isometric view

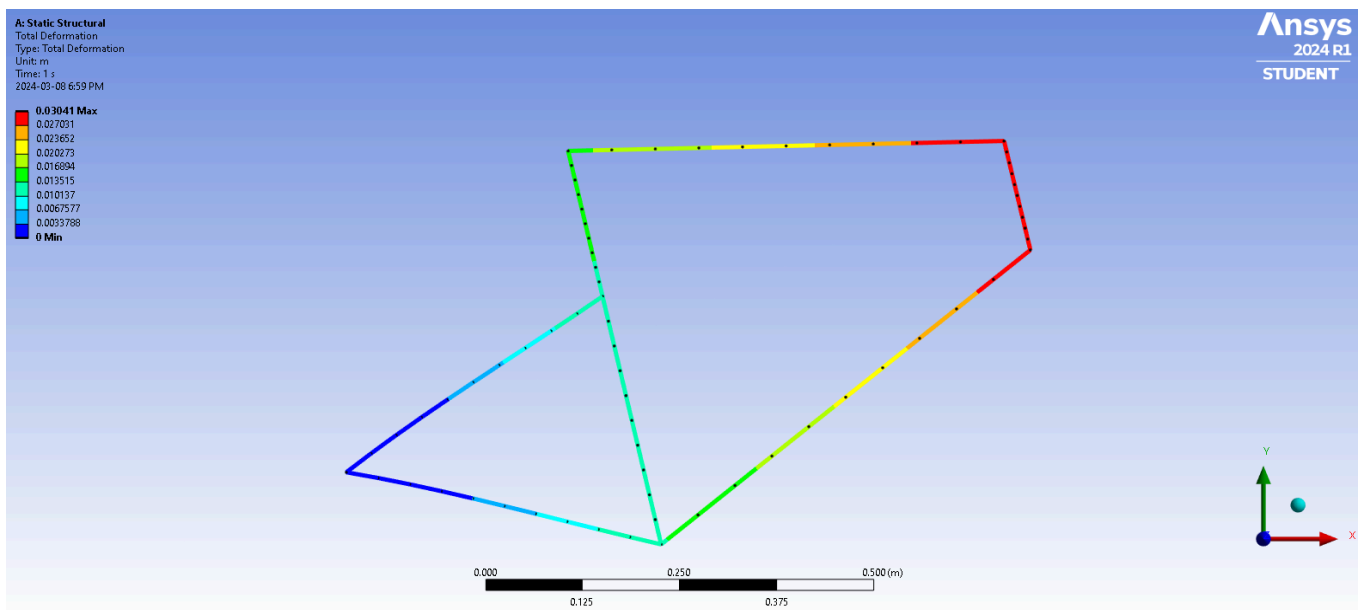


Figure 50: Max Deflection in wire frame side view

Figures 49 and 50 depicted the deformation analysis for the modeled bike frame. Deformation stress analysis allows engineers to evaluate the structural integrity of components or entire systems. By simulating the potential loads and boundary conditions, it predicts how a design will behave under stress. Deformation analysis ensures that the frame can withstand the forces it will encounter during use, such as rider weight, road vibrations, and impacts. In this analysis there are stress concentrations near joints, welds, or critical load-bearing regions. Addressing these weak points early in the design phase prevents unexpected failures. By identifying stress hotspots, designers can make informed decisions about material selection and geometry adjustments. By adjusting material thickness, geometry, or reinforcement, they can enhance performance while minimizing weight.

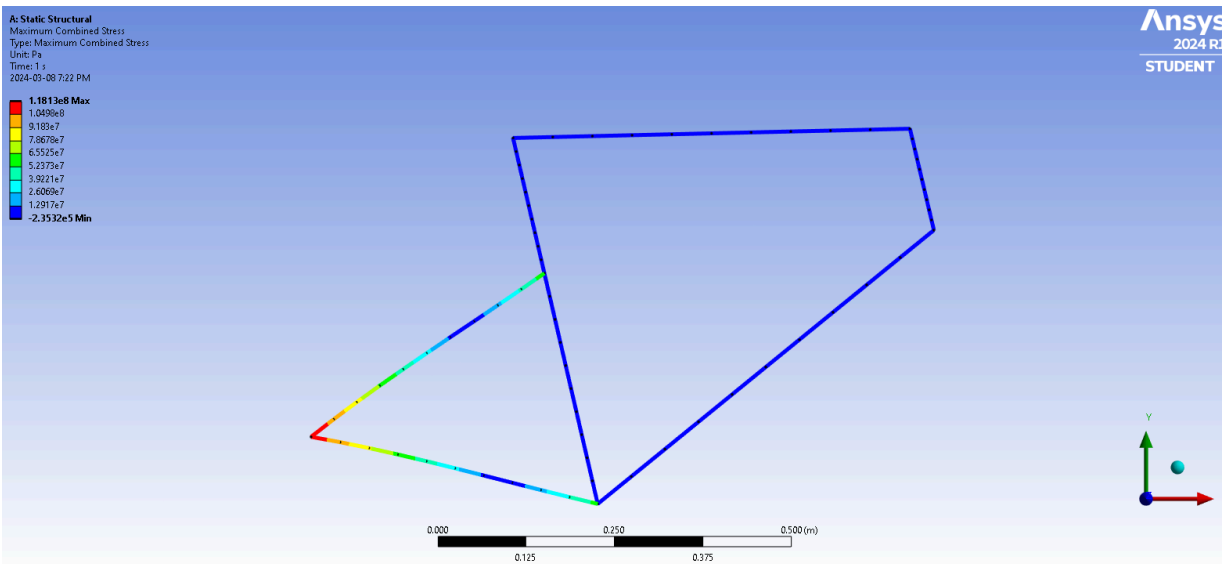


Figure 51: Max Combined Stress Example with wire frame.

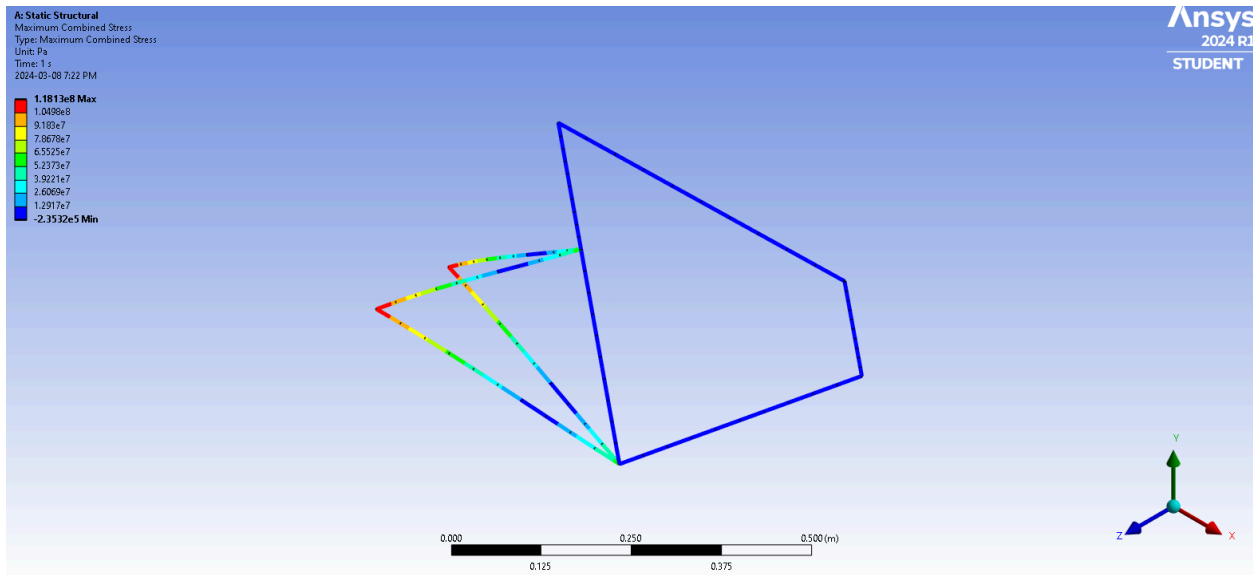


Figure 52: Max Combined Stress Example with wire frame isometric.

Figures 51 and 52 depicted the combined stress analysis for the modeled bike frame. The purpose of this analysis is to evaluate the structural integrity and performance of the bike frame under various (combined) load conditions. The frame is color-mapped to represent different stress levels and the loading condition is when the rider is pedaling the bicycle during competition. Areas with maximum stress are likely highlighted in warm colors (such as red or orange), while regions with minimal stress may appear in cool colors (like blue or green). Engineers can use numerical values for stress, strain, and deformation at specific points on the frame to assess whether it can withstand applied loads without failure. Additionally, the simulation considers various load scenarios, such as forces applied at the pedals, seat, and handlebar, mimicking real-world usage.

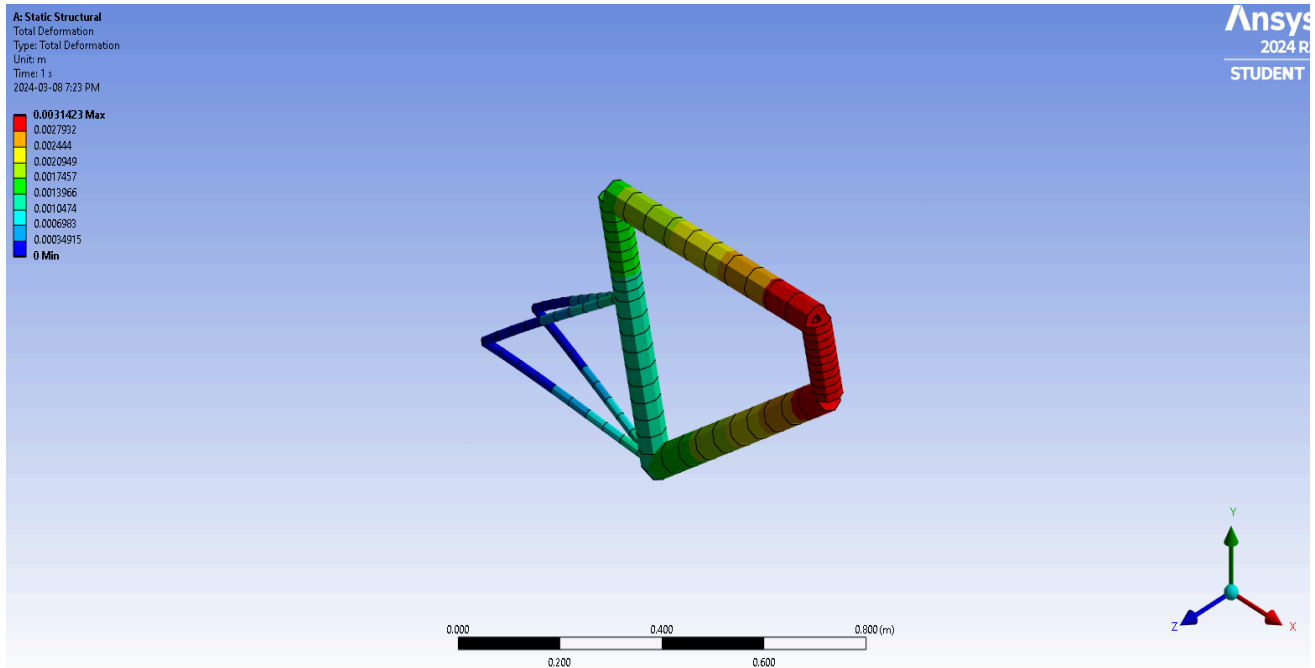


Figure 53: Max deflection example with round cross section

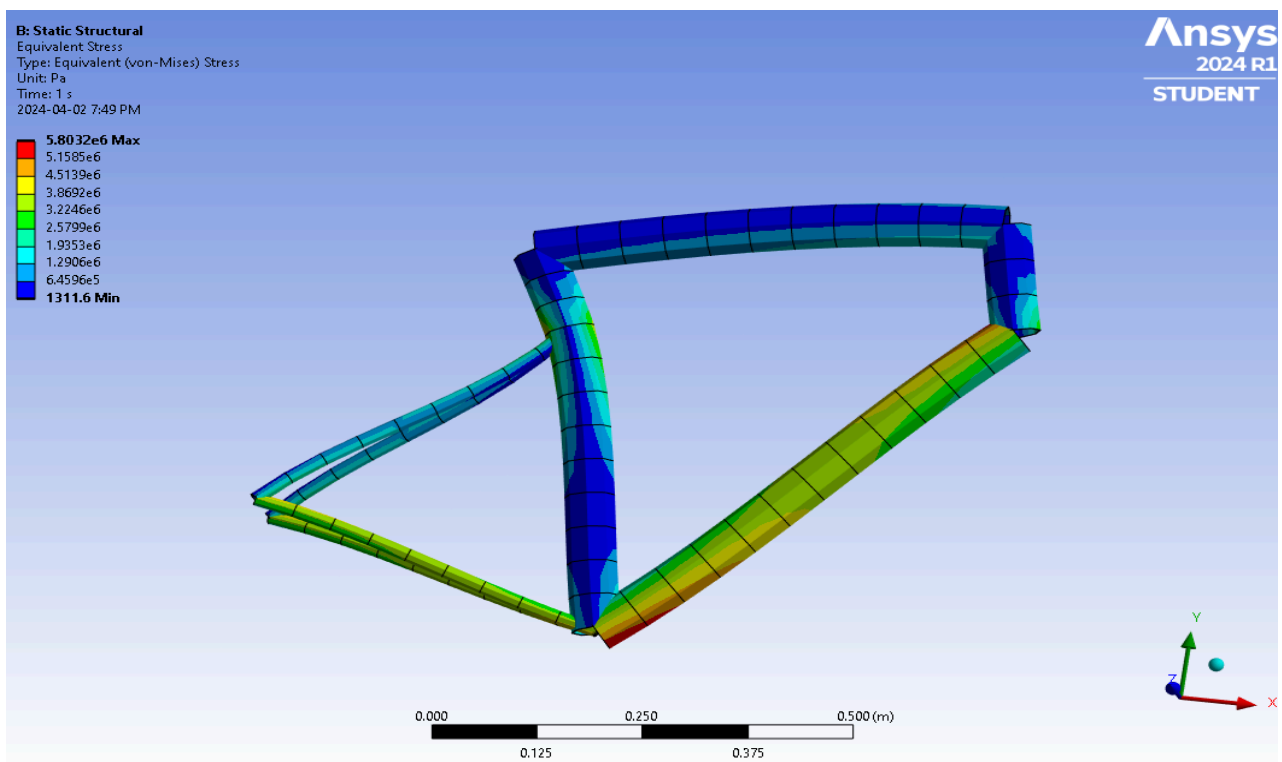


Figure 54: Equivalent (von-Mises) Stress Deflection side view with round cross section.

Figures 53 to 54 depict the von Mises stress analysis for the modeled bike frame. Von Mises stress is a metric used to assess the total amount of stress a material experiences when exposed to intricate loading circumstances, such the ones a bike frame encounters when in use. Von Mises stress takes into account the combined effect of all stress components to estimate the equivalent stress, in contrast to standard stress calculations that concentrate on individual stress components (such as tension or compression). A thorough evaluation of the material's deformation and possible failure locations is provided by the von Mises stress for a bike frame, which experiences a variety of loads and forces during riding. Engineers can determine regions that are vulnerable to fatigue failure or yielding by examining the von Mises stress distribution throughout the frame.

Moreover, Table 9 provides a structured overview of the design strategy for the racing bicycle, highlighting key components such as systems design, detailed design, and prototype construction and evaluation. Each aspect of the design strategy is clearly delineated, facilitating effective communication and coordination among team members and stakeholders involved in the project.

Table 10: Design Strategy based on strategic steps of the conceptual design process.

Design Strategy	Description
1. Systems Design	• Analyze input, process, and output of the prototype. • Identify key performance metrics: speed, aerodynamics, rider comfort. • Guide design process based on performance criteria.
2. Detailed Design	• Use morphological charts to analyze prototype components. • Consider frame geometry, material selection, component integration. • Evaluate optimal choices for performance, manufacturability, cost.
3. Prototype Construction and Evaluation	• Construct a prototype based on finalized design. • Utilize manufacturing techniques: additive manufacturing, carbon fiber layup, CNC machining.

	● Evaluate prototype performance through testing and trials. ● Assess metrics: speed, stability, durability, rider feedback. ● Iterate design to address issues and optimize performance.
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Iteration 3: Updated 8-Member bicycle frame design

In the 3 design iteration, the connection lengths of the seat stays and seat tube were redesigned to understand the relationship between the stresses and the corresponding connection length. In addition, the cross sections of the beams were redesigned based on the critical areas determined in iteration 2. The bicycle frame design for iteration 3 is shown in Figure 55.

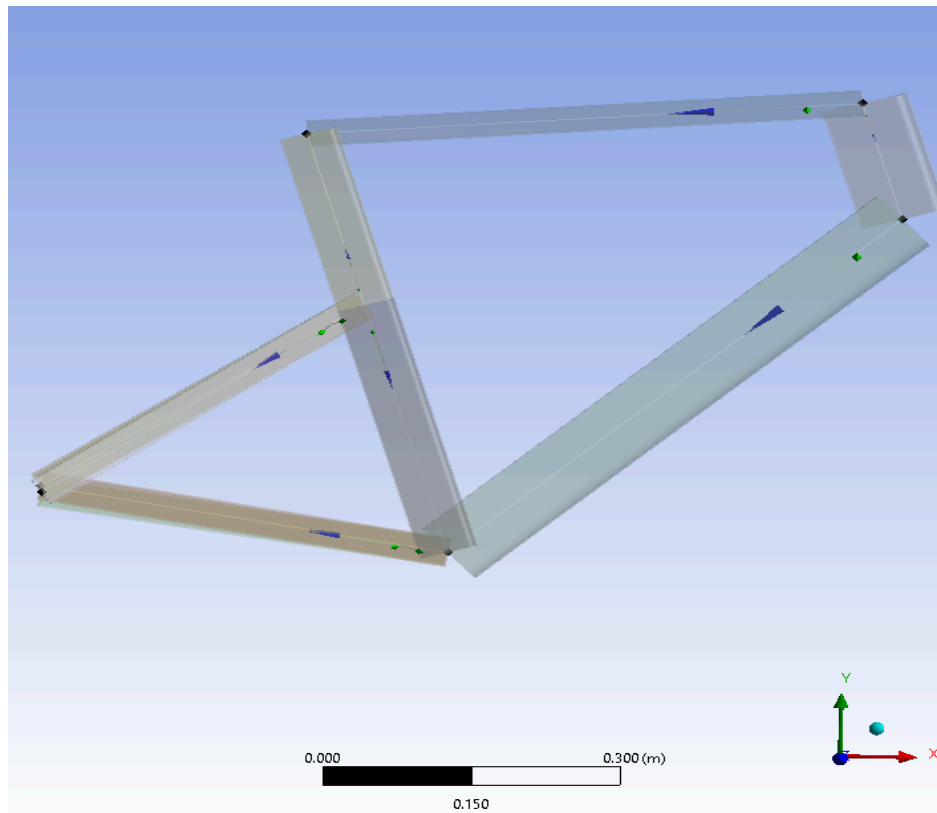


Figure 55: Third Iteration on the 8-Member Bicycle Frame.

The following loading conditions were performed on the third iteration of the bicycle frame based on Alexandros G. Kitselis et al with slight variation [4]:

- Horizontal Loading Scenario
 - Loading force of 600 N applied on each end of the head tube
- Static Start-up
 - Loading force of 1090 N downwards on the end of the bottom bracket
- Steady State Scenario
 - Loading force of 400 N on the bottom bracket, 400 N on the top head tube, 700 N on the seat tube

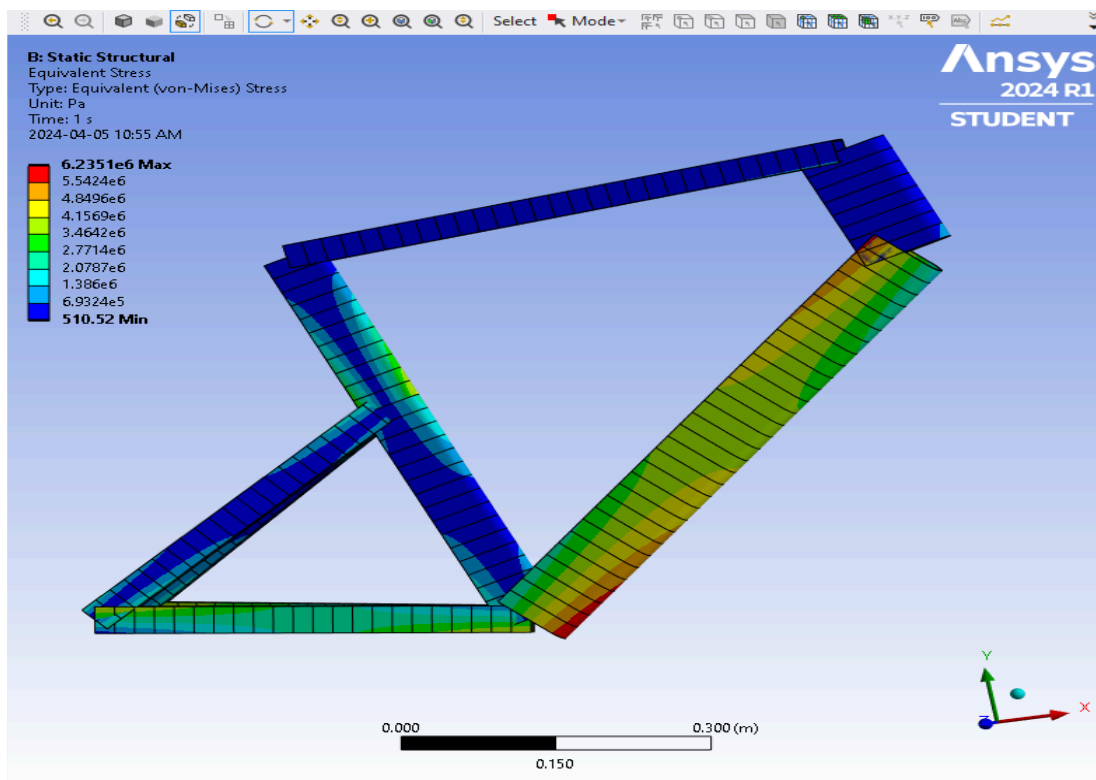


Figure 56: Equivalent (von-Mises) Stress Deflection side view with round cross section with Rear at the lowest position.

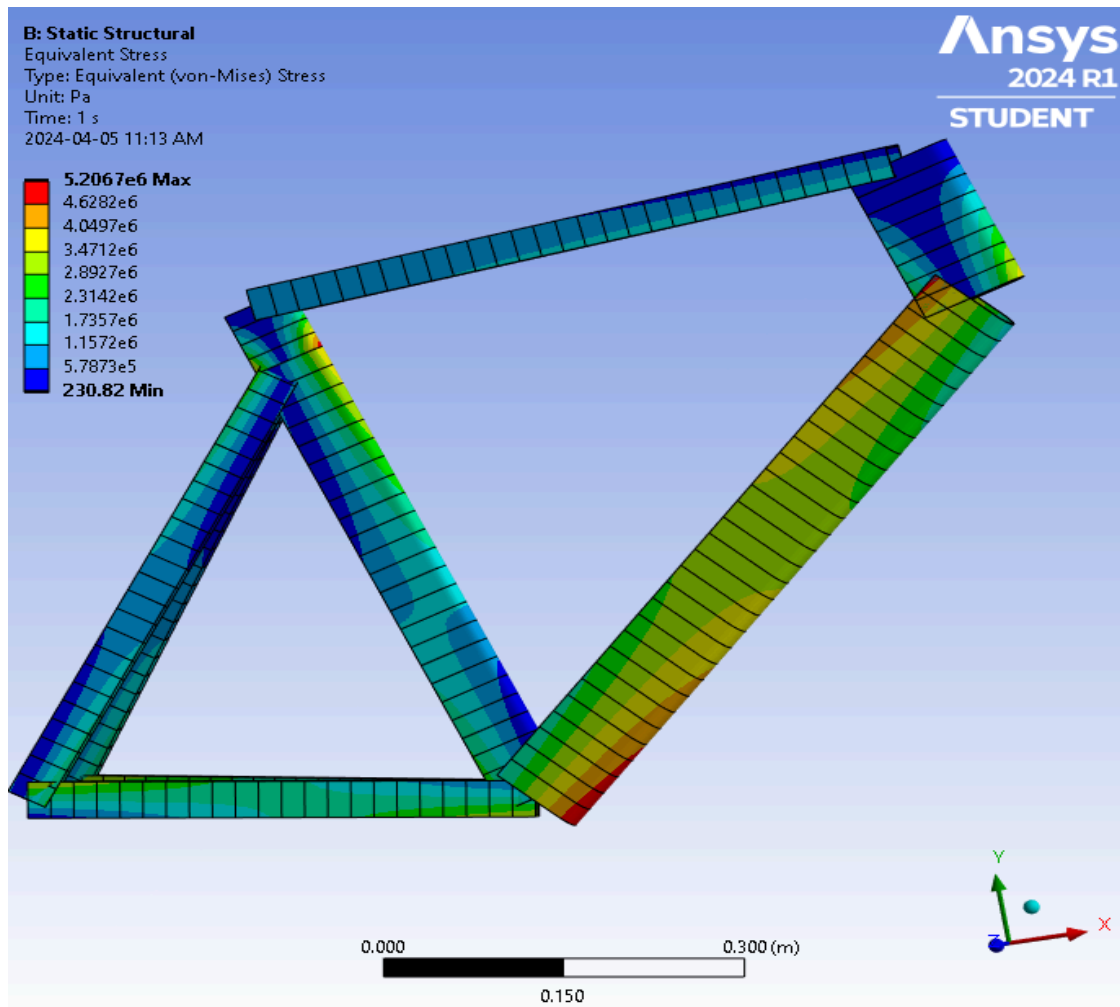


Figure 57: Equivalent (von-Mises) Stress Deflection side view with round cross section with Rear at the highest position.

The effects of the seat stay connection height on the Max Stress and deflection for the various loading conditions was studied to determine the optimal set stay connection height. The graphs for max stress and max deflection are shown in Figures 58, and 59 respectively. From Figure 58, it can be observed that the value of max stress at the three loading conditions was highest when the seat stay connection height is approximately 0.4 m, lowest when the seat stay connection height is approximately 0.45 m. From Figure 59, it can be observed that as the height of the seat stay connection increases, the values of the max deflection decreases.

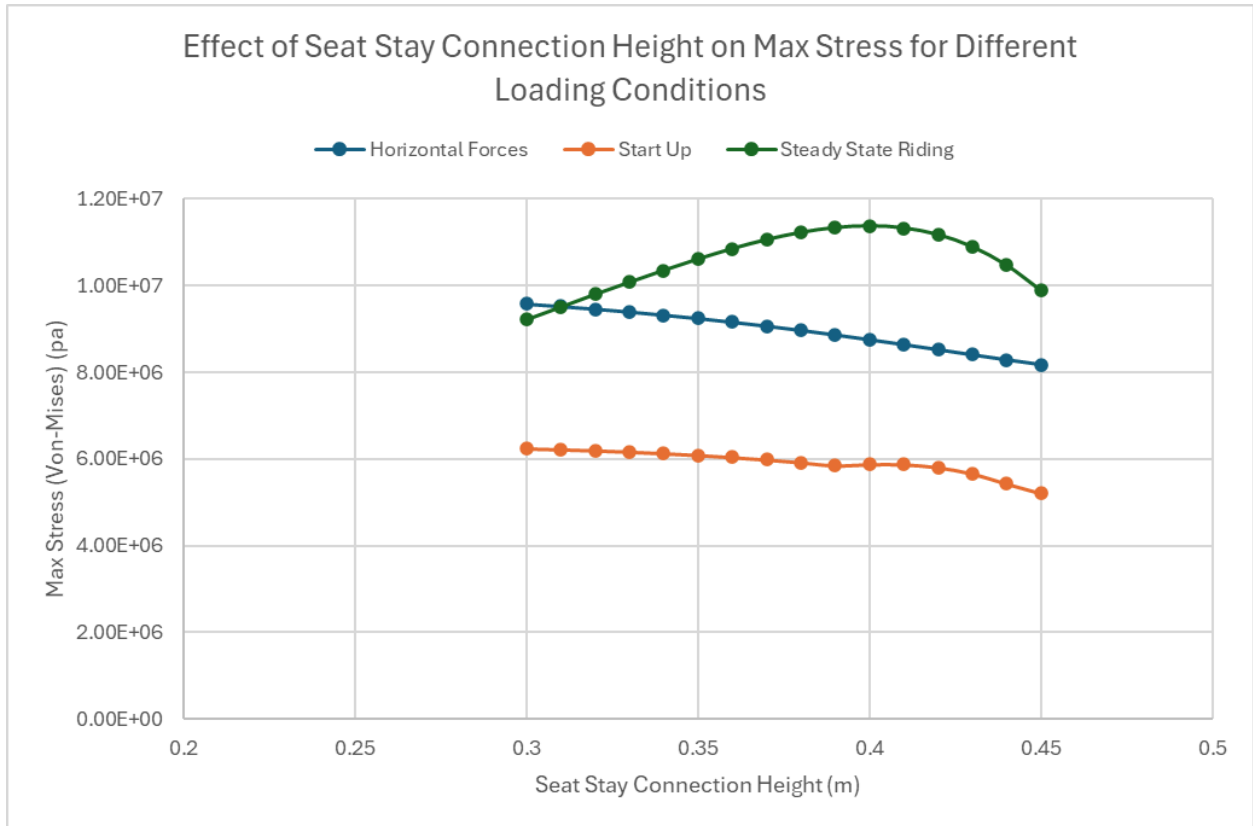


Figure 58: Effect of Seat Stay Connection Height on Max Stress for Loadings.

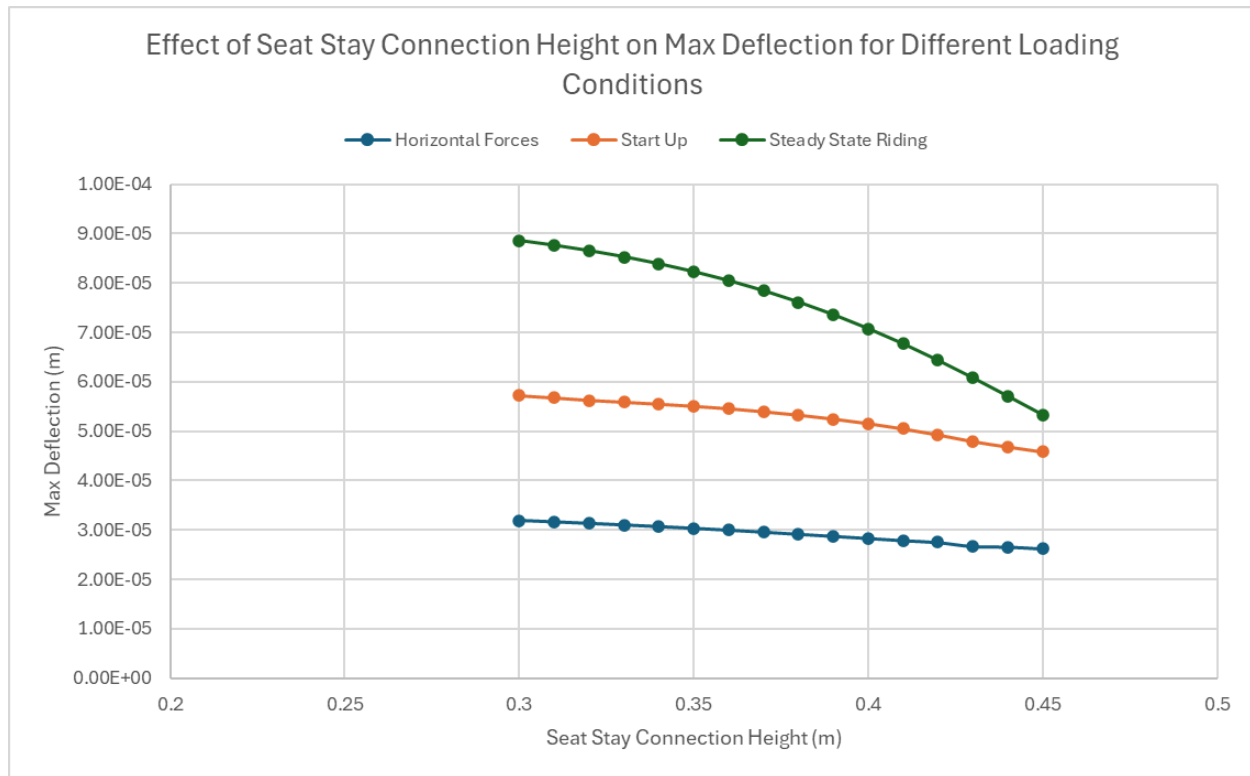


Figure 59: Effect of Seat Stay Height on Max Deflection for Loadings.

3.2. Systems Design

Essentially, constructing a systems design model is an important factor in the engineering design process as it serves as a foundational framework for conceptualizing, organizing, and optimizing complex engineering systems analysis. Moreover, by systematically analyzing the inputs, processes, and outputs of a system, engineers can gain a holistic understanding of the functionality and performance requirements of the final product. Ultimately, a well-executed systems design phase lays the groundwork for efficient implementation, effective problem-solving, and successful project outcomes across a wide range of engineering disciplines and applications. Furthermore, a simplified version of a systems model for tackling the problem scenario of reconceptualizing the racing bicycle is illustrated in Figure 54. Furthermore, a more detailed version of the system's design is presented in Figure 55.

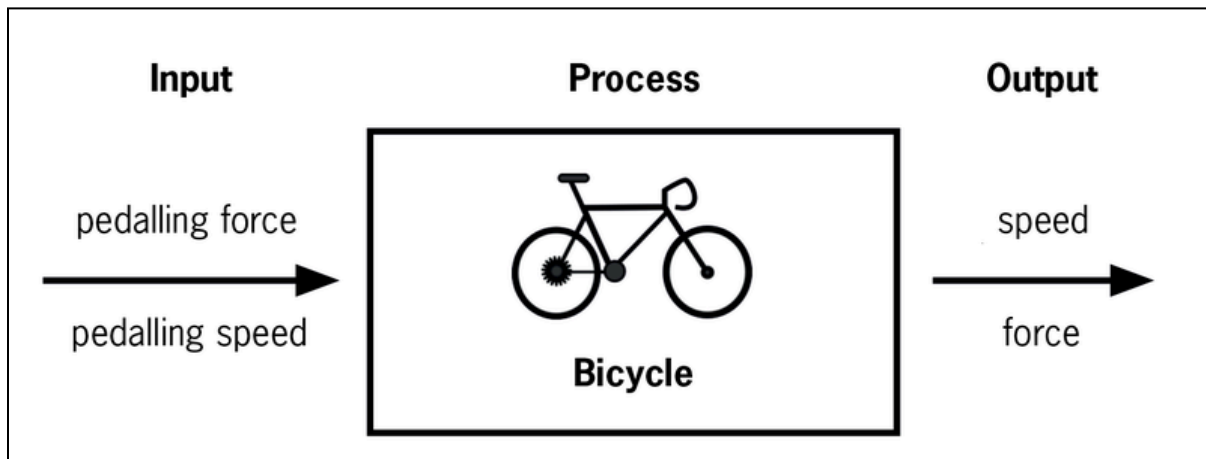


Figure 60: Simplified System Design of the Bicycle [3].

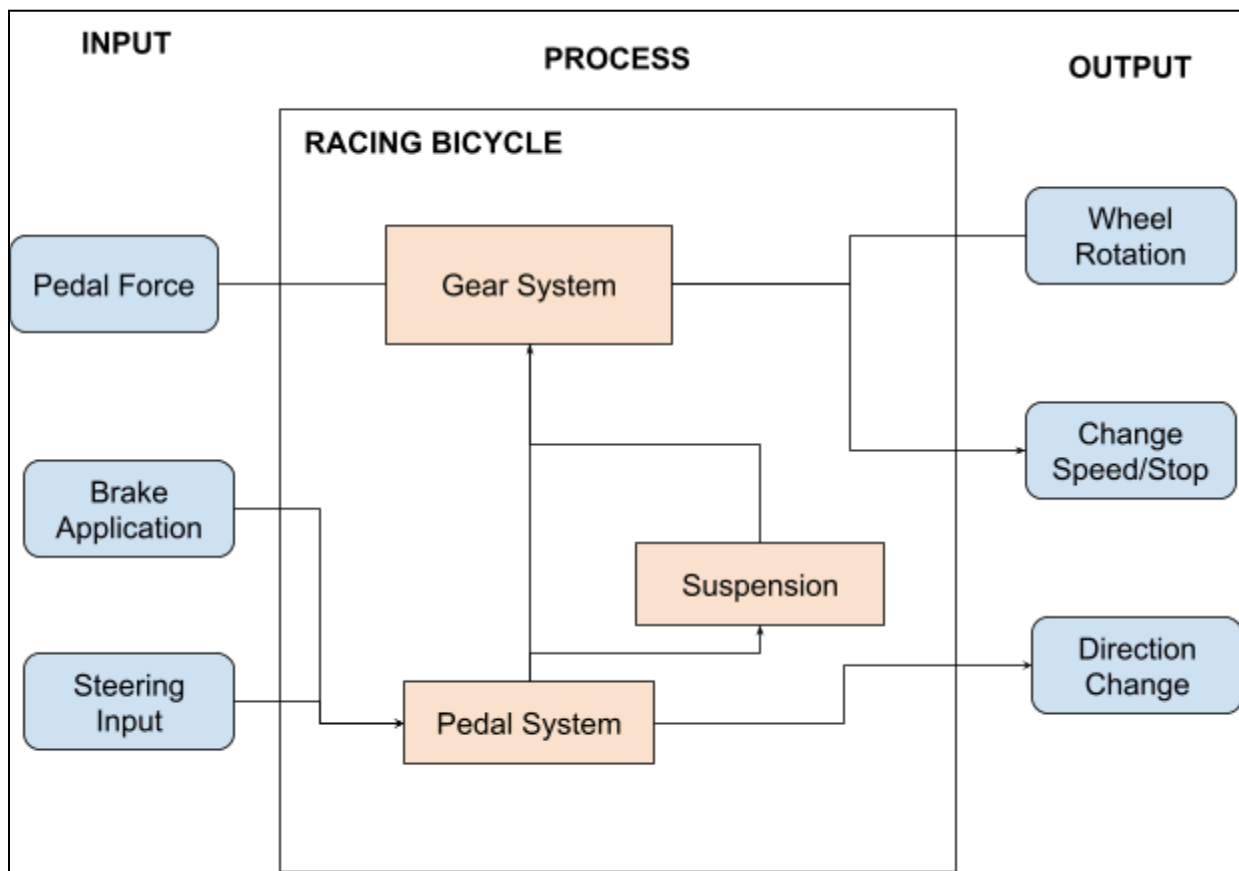


Figure 61: Detailed Version (Black Box Diagram) of the System Design of the Bicycle [3].

3.3. Detailed Design

According to Table 10, tests were conducted on different shapes for each part of the bicycle frame to find the best cross-sectional design. As a result, a cylindrical 40 mm top tube was selected due to its consistent performance across all tests. For the down tube, which experiences significant airflow, we combined three oval-shaped cross-sections: the 39×79 for aerodynamics and low mass, along with the 51×79 for stiffness. The seat tube, a critical element, was chosen based on structural tests and CFD analysis. Despite its higher mass, we opted for complex 1 due to better aerodynamic behavior. The seat stay, responsible for vibration absorption, features an oval $30 \text{ mm} \times 10 \text{ mm}$ shape. Lastly, the chainstays strike a balance between reduced drag and necessary stiffness by combining an oval $20 \text{ mm} \times 60 \text{ mm}$ outside diameter with a rectangular inside diameter. Overall, each part was optimized for performance and stability. Therefore, as per the requirements for the tube diameter sizes and types, the CAD Drawing is produced in Figure 3 below with references to the detailed view of each tube diameters.

Table 11: Evaluation of each cross-section for the bike frame parts [4]

Frame Parts	Cross-Section	Displacement	Mass	Weighted Drag	Cost
Top Tube	cylindrical	✓	✓	✓	✓
	rectangular	✓	✗	✗	✓
	triangular	✗	✓	✓	✗
	complex	✗	✓	✗	✗
Down Tube	oval	✓	✓	✓✓	✓
	cylindrical	✗	✓	✗	✓
	rectangular	✗	✓	✗	✓
	complex1	✓	✓	✓	✗

	complex2	✓	✗ ✗	✓	✗
Seat Tube	complex1	✓	✗	✓	✗
	complex2	✓	✓	✓	✗
	Cylindrical (53 mm)	✓	✗	✗	✓
	Cylindrical (35 mm)	✗ ✗	✓	✓	✓
Seat Stays	Oval (10x30 mm)	✗	✓	✓✓	✓
	Oval (17x31 mm)	✓	✗	✗	✓
	rectangular	✓✓	✗	✓	✗
	cylindrical	✗	✓✓	✗	✓
Chain Stays	Oval (15 x 45 mm)	✗	✓	✓✓	✓
	Oval (20 x 60mm)	✓	✓	✓	✓
	Rectangular (15 x 45 mm)	✗	✓	✓	✓
	Rectangular (20 x 60 mm)	✓✓	✗ ✗	✗ ✗	✓
	Cylindrical	✗ ✗ ✗	✓✓	✓✓	✓

- ✓ indicates a good behavior–result
- ✓✓ indicates a very good behavior–result
- ✗ indicates a bad behavior–result
- ✗ ✗ indicates a very bad behavior–result
- ✗ ✗ ✗ indicates an unacceptable result.

Furthermore, based on the literature review and the detailed design of the bicycle from Table 10, Table 11 presents a morphological chart for the detailed design matrix. The key selection for the finalized version is highlighted in yellow.

Table 12: Morphological Chart

Bike Components	Options			
	1	2	3	4
Frame	High Seat Diamond	Low Seat Diamond	HSLR	
Material	Aluminum	CFRP	Steel	Titanium
Transmission System	Chain	Toothed Belts	Cardan Shaft	
Wheel Tire	Mountain Tire	Gravel Tire	Road Tire	Commuter Tire

Explanation of Selection

1. Frame:

- Option 3 (HSLR): This frame design strikes a balance between comfort and control. The higher seat allows for an upright riding position, reducing strain on the back.
- Explanation: In racing mountain bikes, durability and precise handling are essential. The High Seat Diamond frame provides a solid foundation for aggressive riding on challenging terrain. The low rear diamond frame provides stability and ease of mounting/dismounting. Moreover, This frame design is known for its strength and rigidity, offering stability and control crucial for mountain biking.

2. Material:

- Option 2 (CFRP): Carbon fiber is the top choice for racing mountain bikes. Here's why:
 - Lightweight: Carbon frames are exceptionally light, aiding acceleration and agility.

- Strength: Carbon fiber provides a high strength-to-weight ratio, crucial for handling rough trails.
- Vibration Damping: Carbon dampens vibrations, enhancing rider comfort during long races.
- Explanation: A carbon fiber frame ensures optimal performance, especially when navigating technical descents and climbs.

3. Transmission System:

- Option 3 (Cardan Shaft): While not common in mountain bikes, the Cardan Shaft offers advantages:
 - Durability: Cardan shafts require minimal maintenance and withstand harsh conditions.
 - Efficiency: Power transfer is efficient, crucial for sustained efforts during races.
- Explanation: In long, rigorous races, the Cardan Shaft's durability and low maintenance can provide an edge over traditional chains or toothed belts.

4. Wheel Tires:

- Option 1 (Mountain Tire): Specifically designed for grip and control on uneven terrains.
 - Tread Pattern: Mountain tires have aggressive tread patterns for traction on dirt, rocks, and roots.
 - Wide Profile: Wider tires provide stability during technical descents.
- Explanation: Mountain tires ensure confident handling and traction, allowing one to push the limits in races.

3.4. Prototype Construction and Evaluation

Based on Figure 56, a third iteration of the bicycle frame prototype is finalized. In this design, smooth weldments on a bike frame provide various benefits, such as improved aesthetics and less aerodynamic drag, which increases performance. By reducing stress concentrations and possible areas of failure, smooth weldments also enhance structural integrity and durability. Furthermore, the lack of visible welds can produce a smooth and seamless look, which appeals to

customers who value craftsmanship and aesthetics while making purchases. However, because seamless junctions require specific welding procedures and post-processing treatments, this approach may result in greater manufacturing costs and technical complexity.

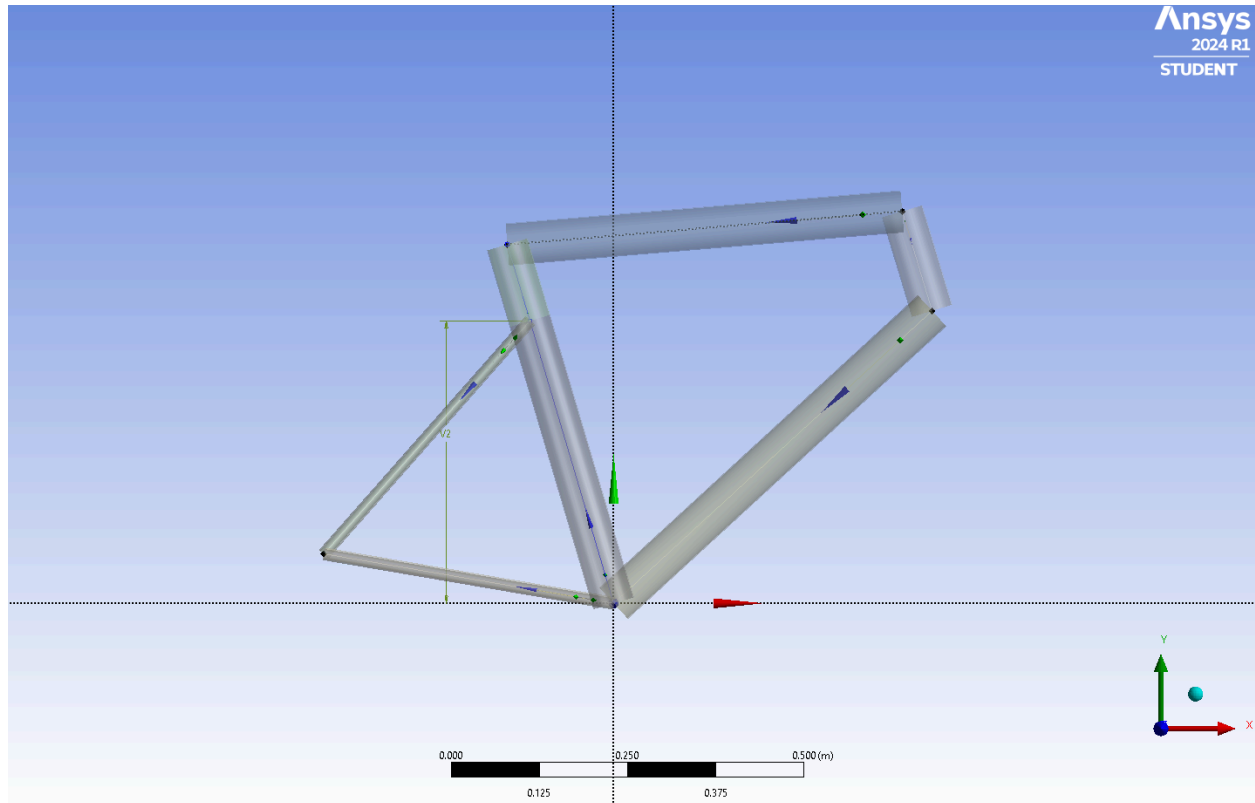


Figure 62: Final frame model constructed on ANSYS.

However, if a bike frame is designed without weldments, there is no need for welding at all, which streamlines the manufacturing process and might save production costs. Since it removes the possibility of heat-affected zones and welding-related distortion, this method may also lead to a reduced frame weight and better fatigue resistance. But smooth weldments provide frames their aesthetic appeal and design flexibility; this could limit the customization choices and attraction to consumers for frames without weldments. Furthermore, employing sophisticated fabrication techniques like bonding or additive manufacturing to provide seamless junctions devoid of weldments may be necessary, which could add to the complexity and present quality control issues.

Therefore, the decision to design a bicycle frame with regards to weldments involves analyzing the factors of aesthetics, performance, manufacturing, cost, and technical feasibility based on the specific requirements goals. Since weldments have a higher advantage weightage than without, the final prototype of the bike frame is decided to include welds. The final prototype with rendering is presented on the Figure 58 below.



Figure 63: Rendering of the Final Version.

4. DISCUSSIONS AND CONCLUSIONS

4.1. Comparison with Existing Designs

A detailed comparison is drawn between the newly developed racing bicycle frame and existing designs in the market or literature. With comparisons of the key features, performance metrics, and design attributes of both the new and existing frames, a comprehensive assessment of the innovations and improvements introduced in the new design is presented. This comparison highlights areas of improvement such as improved aerodynamics, stiffness-to-weight ratio, or superior manufacturability. Additionally, any shortcomings or limitations of the new design relative to existing solutions are acknowledged and discussed, providing valuable insights for future iterations and refinements. Through this comparative analysis, the strengths and weaknesses of the new racing bicycle frame are contextualized within the broader landscape of contemporary design practices, contributing to a nuanced understanding of its potential impact and significance in the cycling industry.

4.2. Conclusions and Recommendations

In conclusion, the key findings proven during the design process throughout this report has provided an overview of the project outcomes according to their objective. Conclusions drawn from the analysis of the performance data, simulation results, and user feedback has shed light on the effectiveness of the racing bicycle frame design in meeting the objectives. Additionally, recommendations are proposed for future research, development, and refinement of the design methodology, including suggestions for further optimization, validation, and real-world testing. By encapsulating the project's achievements and offering strategic guidance for future endeavors, this section aims to inform and inspire continued innovation in the field of racing bicycle design.

REFERENCES

- [1] Clarification guide of the UCI technical regulations,
https://assets.ctfassets.net/76117gh5x5an/6cmP3f8KY4mTf16hAYCduP/5e606e6e9a6ca688a35e625a25947a3f/Clarification_Guide_of_the_UCI_Technical_Regulation_-_20230101_-_ENG.pdf
- [2] Current Perspectives of Cross-Country Mountain Biking: Physiological and Mechanical Aspects, Evolution of Bikes, Accidents and Injuries. Available:
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565958/>
- [3] Bevel gears, bicycles and systems diagrams. Available:
<http://www.mstworkbooks.co.za/technology/gr8/gr8-technology-14.html>
- [4] A. G. Kitselis, C. S. Nikolakea, and D. E. Manolakos, "The design process of an optimized road racing bicycle frame," *Machines*, vol. 10, no. 2, p. 149, Feb. 2022.
doi:10.3390/machines10020149  [The_Design_Process_of_an_Optim.pdf](#)
- [5] Sarath. P, A. Deepak, N. S. Daniel, H. H, and Jinuchandran, "Stress Analysis of Bicycle Frame using Different Materials by FEA," *GRD Journal for Engineering*, vol. 6, no. 7, Jun. 2021. [Online]. Available:
<https://grdjournals.com/uploads/article/GRDJE/V06/I07/0016/GRDJEV06I070016.pdf>
- [6] Akhyar *et al*, "Structural Simulations of Bicycle Frame Behaviour under Various Load Conditions," *Mater. Sci. Forum*, vol. 961, pp. 137-147, 2019. Available:
<http://ezproxy.lib.torontomu.ca/login?url=https://www.proquest.com/scholarly-journals/structural-simulations-bicycle-frame-behaviour/docview/2252435217/se-2>. DOI:
<https://doi.org/10.4028/www.scientific.net/MSF.961.137>.
- [7] "The all-in-one workspace for your notes, tasks, wikis, and databases.," Notion,
<https://hced.notion.site/d9bf444d847a43e9b7bfa9efada98f8f?v=c9f54b0281b644ca9410d3aebd2bc78a> (accessed Apr. 2, 2024).
- [8] E. Levarda, "Bicycle Transmissions," Shieldsquare Captcha,
<https://iopscience.iop.org/article/10.1088/1757-899X/444/5/052013/pdf>.
- [9] S. Inc, "How to choose bike tires [size + width] the bike shoppe," The Bike Shoppe,
<https://www.thebikeshoppe.com/articles/how-to-choose-bike-tires-pg1496.htm>

- [10] K. V. Smith and T. Marshall, "Competitive Cycling in Canada," The Canadian Encyclopedia, <https://www.thecanadianencyclopedia.ca/en/article/cycling> (accessed Apr. 4, 2024).
- [11] J. Rutberg, "Ultimate Guide to getting started in road racing," USA Cycling, <https://usacycling.org/article/ultimate-guide-to-getting-started-in-road-racing> (accessed Apr. 4, 2024).
- [12] "Various frames," Vintage Mountain Bike Workshop, <http://www.vintagemtbworkshop.com/various-frames.html>
- [13] "2025 transition bikes," The Path Bike Shop, <https://www.thepathbikeshop.com/collections/2025-transition-bikes>
- [14] "Pinarello Prince FX Road Frameset - Bikes," Competitive Cyclist, <https://www.competitivecyclist.com/pinarello-prince-fx-road-frameset>

APPENDIX A - DETAILED DRAWINGS



Figure 64: Image of a Traditional Bicycle.

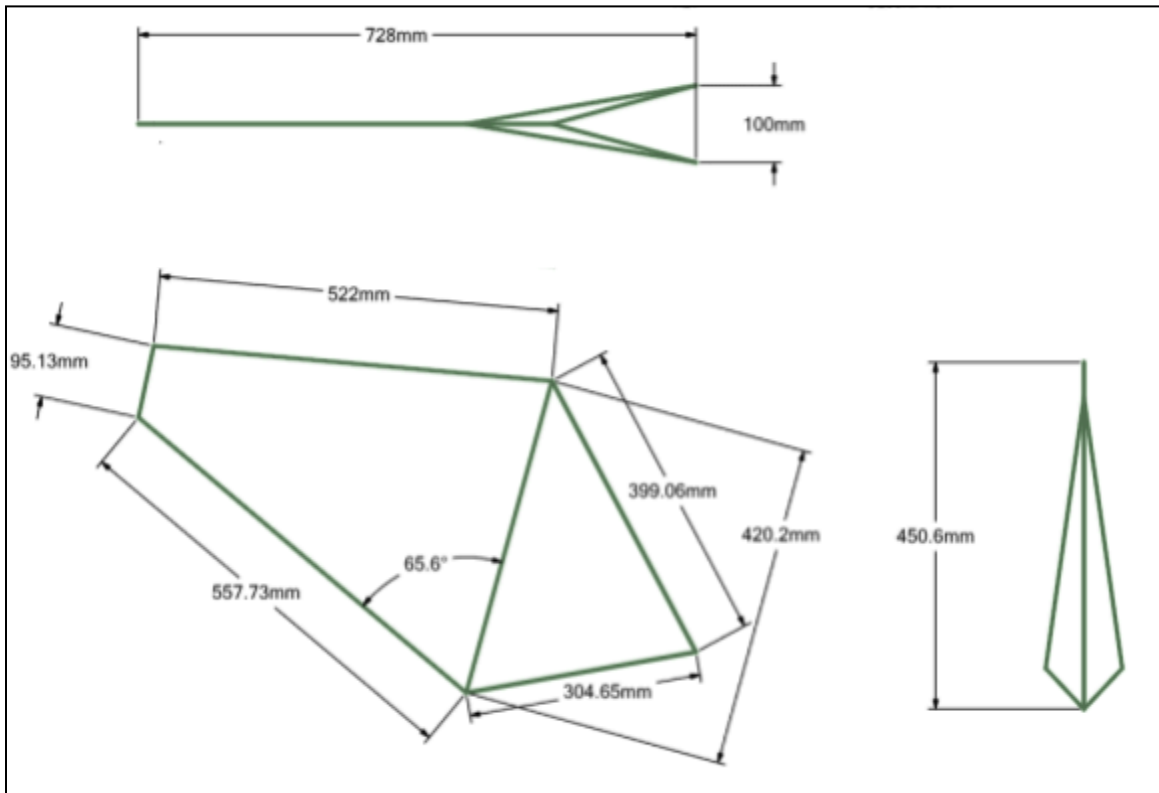


Figure 65: First Iteration Drawing of the Bike Frame (Traditional Mountain Bike).

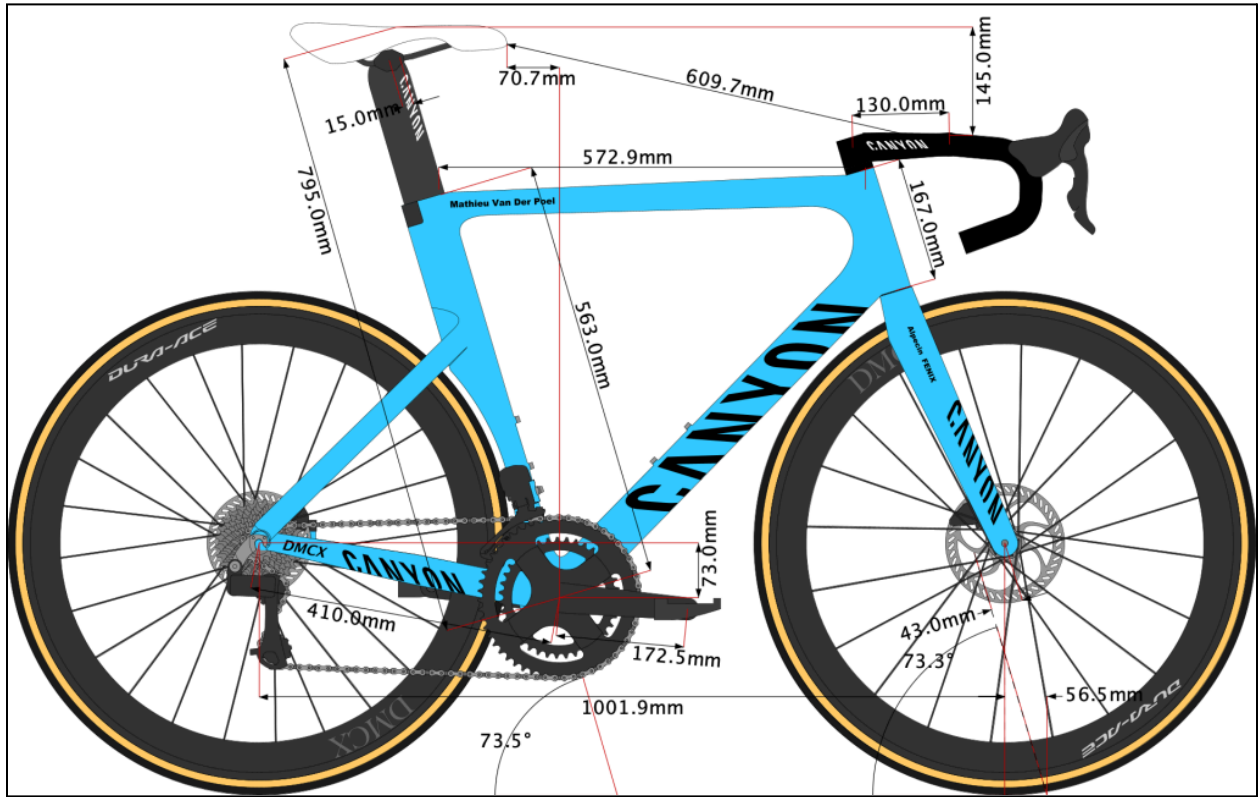


Figure 66: Reference Racing Bike Canyon Aeroad CF SL 7 Disc [5]

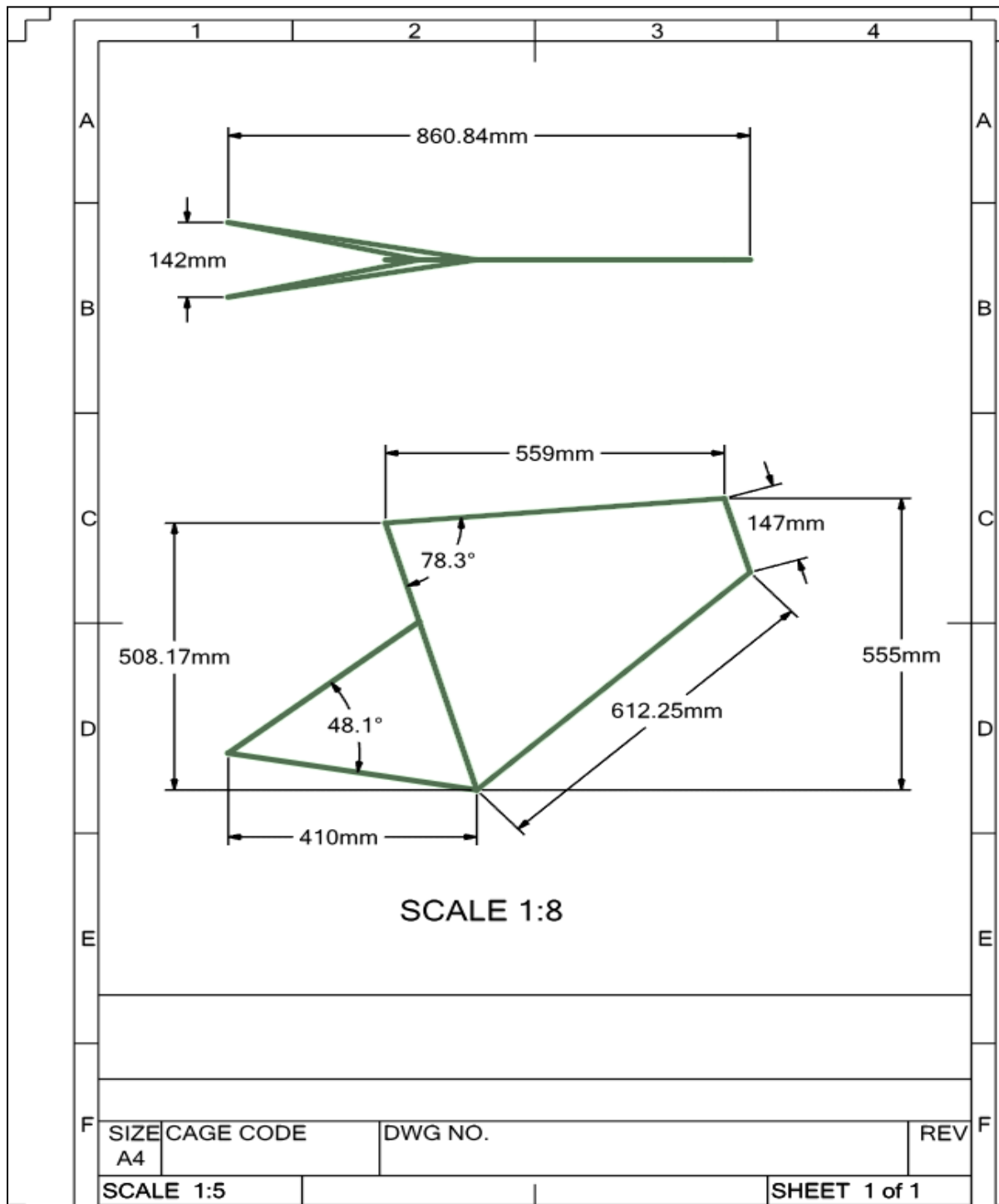


Figure 67: 2nd iteration of bike frame design using existing reference .

APPENDIX B - LOADING CONDITION ANALYSIS

Table 12: Loading Condition Excel Chart

h	Horizontal Stress	Horizontal Deflection	Start Stress	start Deflection	Steady Stress	Steady Deflection
0.3	9.58E+06	3.19E-05	6.24E+06	5.72E-05	9.22E+06	8.86E-05
0.31	9.52E+06	3.16E-05	6.21E+06	5.68E-05	9.51E+06	8.76E-05
0.32	9.46E+06	3.13E-05	6.19E+06	5.63E-05	9.80E+06	8.65E-05
0.33	9.39E+06	3.10E-05	6.16E+06	5.59E-05	1.01E+07	8.53E-05
0.34	9.32E+06	3.07E-05	6.13E+06	5.55E-05	1.04E+07	8.39E-05
0.35	9.24E+06	3.04E-05	6.08E+06	5.51E-05	1.06E+07	8.23E-05
0.36	9.16E+06	3.00E-05	6.04E+06	5.46E-05	1.08E+07	8.05E-05
0.37	9.07E+06	2.96E-05	5.98E+06	5.40E-05	1.11E+07	7.84E-05
0.38	8.97E+06	2.92E-05	5.92E+06	5.33E-05	1.12E+07	7.61E-05
0.39	8.86E+06	2.87E-05	5.85E+06	5.25E-05	1.13E+07	7.36E-05
0.4	8.75E+06	2.83E-05	5.87E+06	5.16E-05	1.14E+07	7.08E-05
0.41	8.64E+06	2.79E-05	5.87E+06	5.05E-05	1.13E+07	6.77E-05
0.42	8.52E+06	2.75E-05	5.80E+06	4.93E-05	1.12E+07	6.44E-05
0.43	8.40E+06	2.67E-05	5.66E+06	4.79E-05	1.09E+07	6.08E-05
0.44	8.29E+06	2.66E-05	5.43E+06	4.68E-05	1.05E+07	5.71E-05
0.45	8.18E+06	2.62E-05	5.21E+06	4.58E-05	9.90E+06	5.34E-05

APPENDIX C - DRAWINGS FOR CROSS SECTIONS

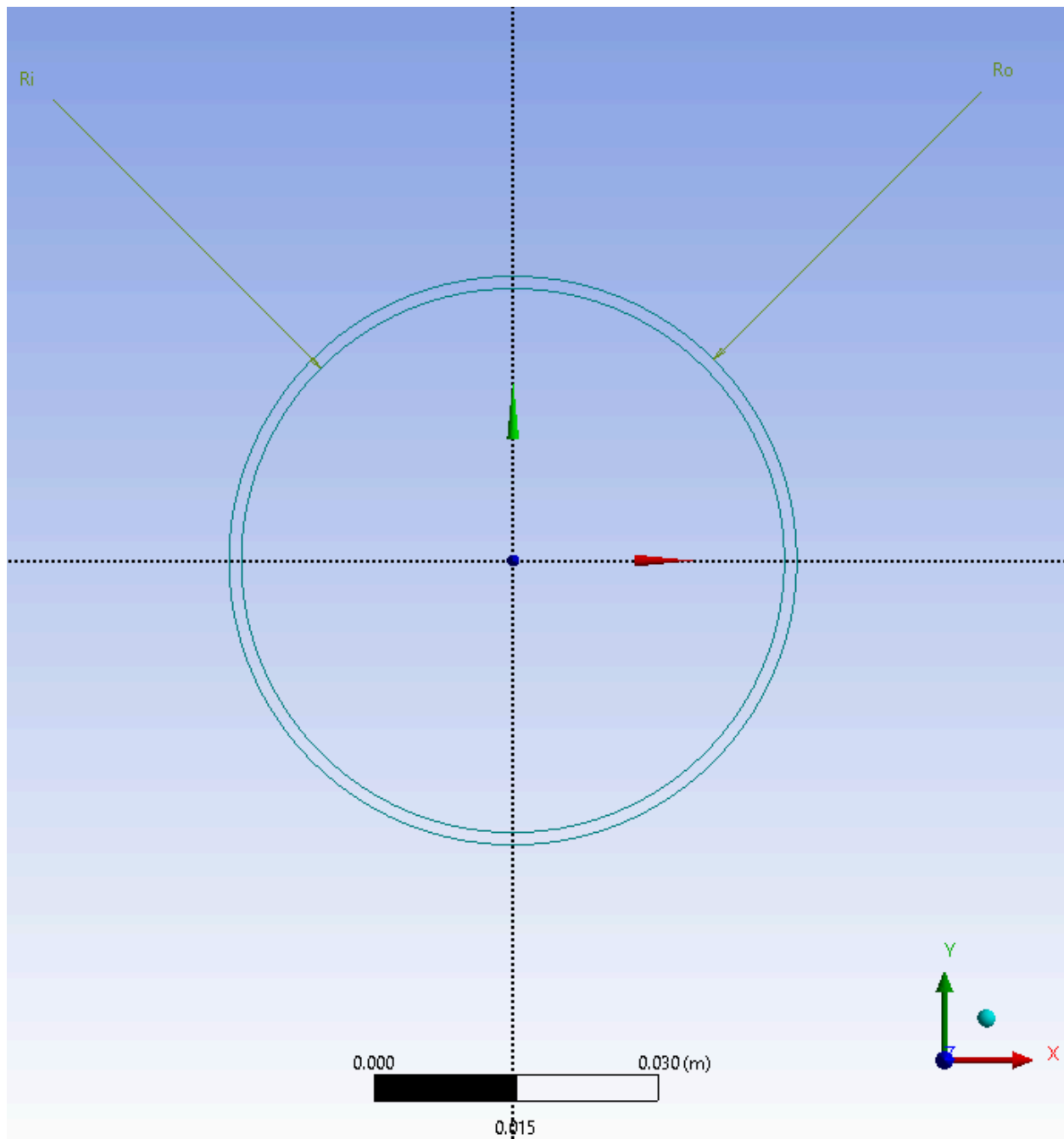


Figure 68: Large Circular Tube

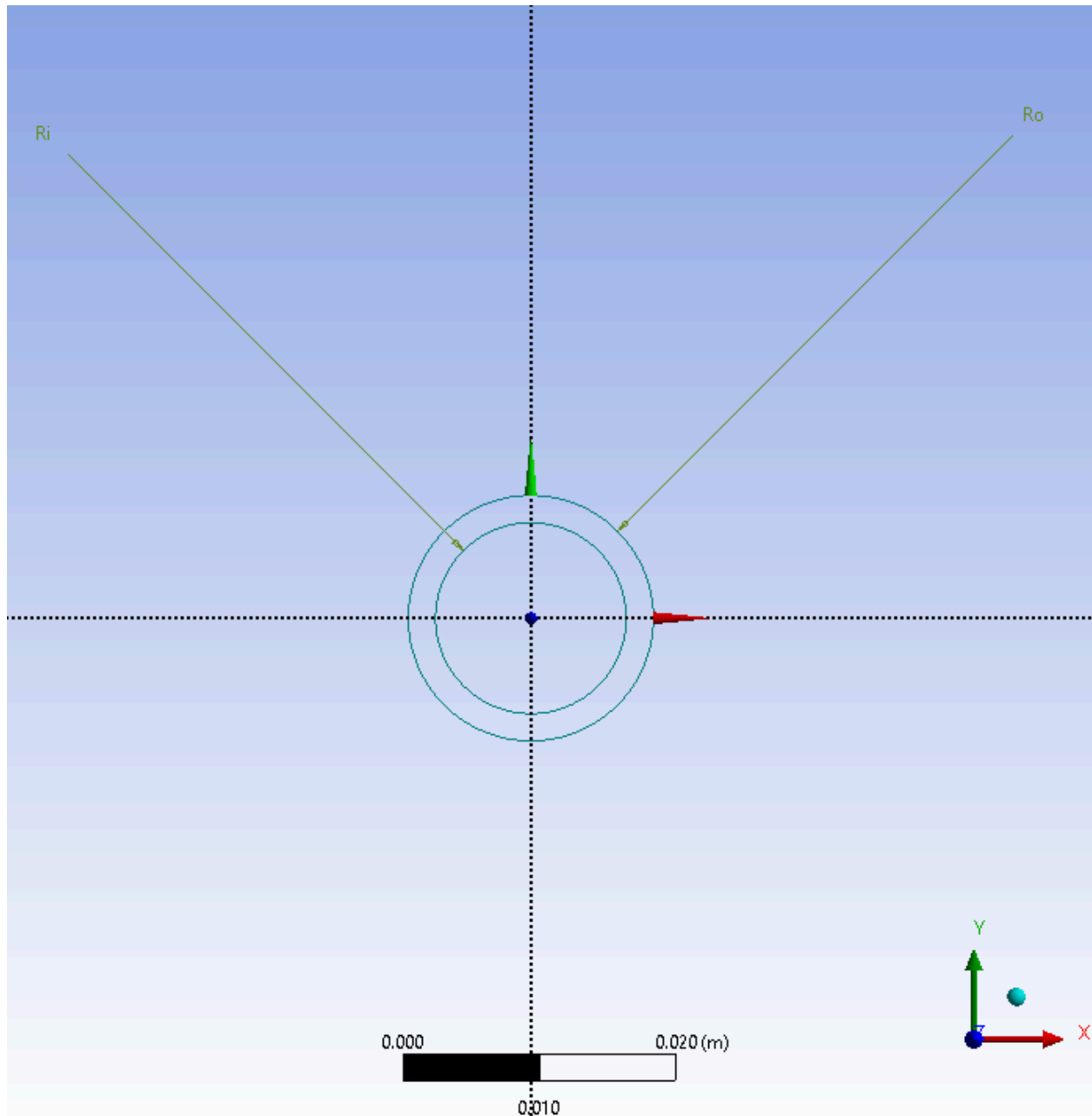


Figure 69: Small Circular Tube

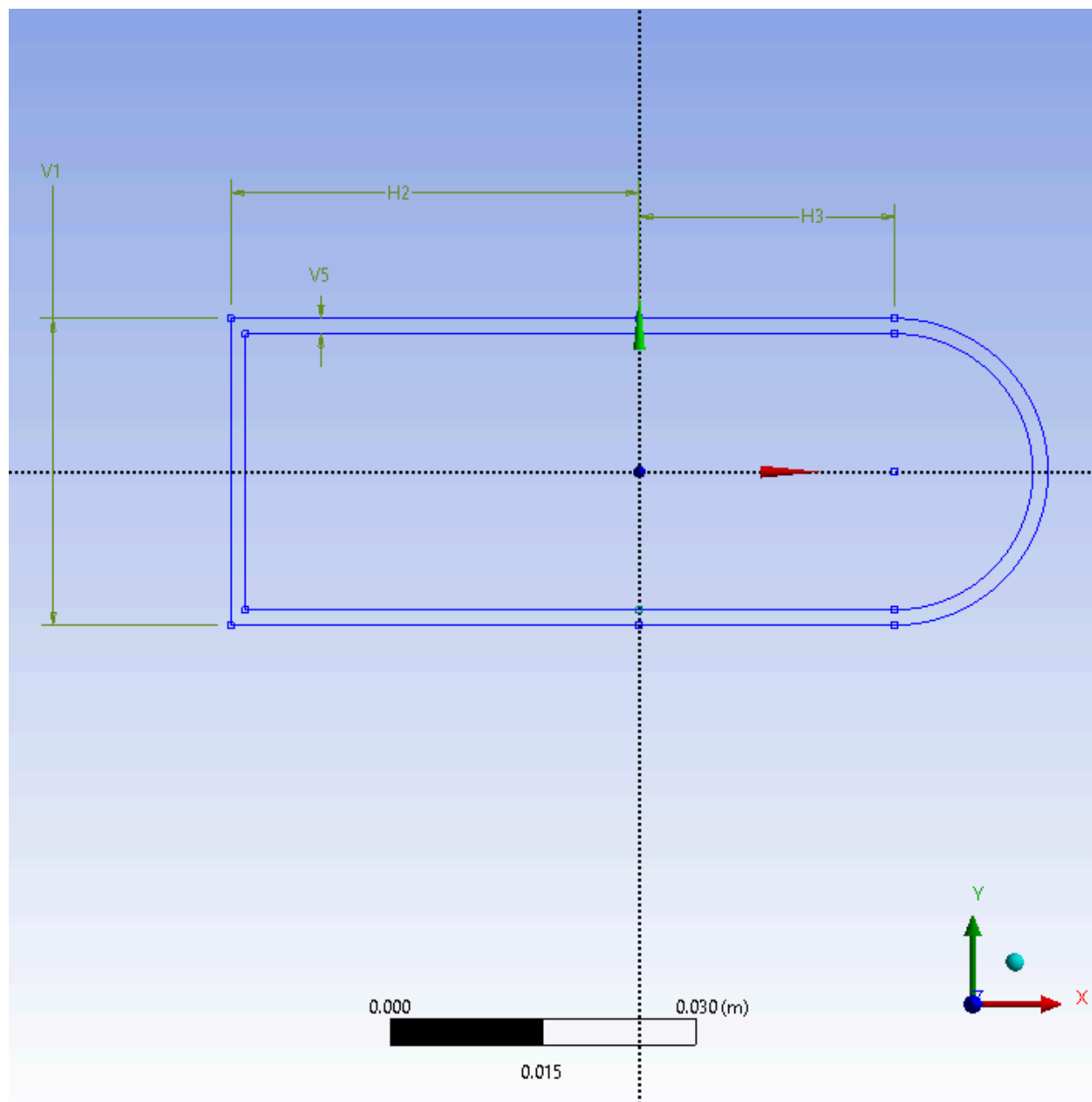


Figure 70: Down Tube.

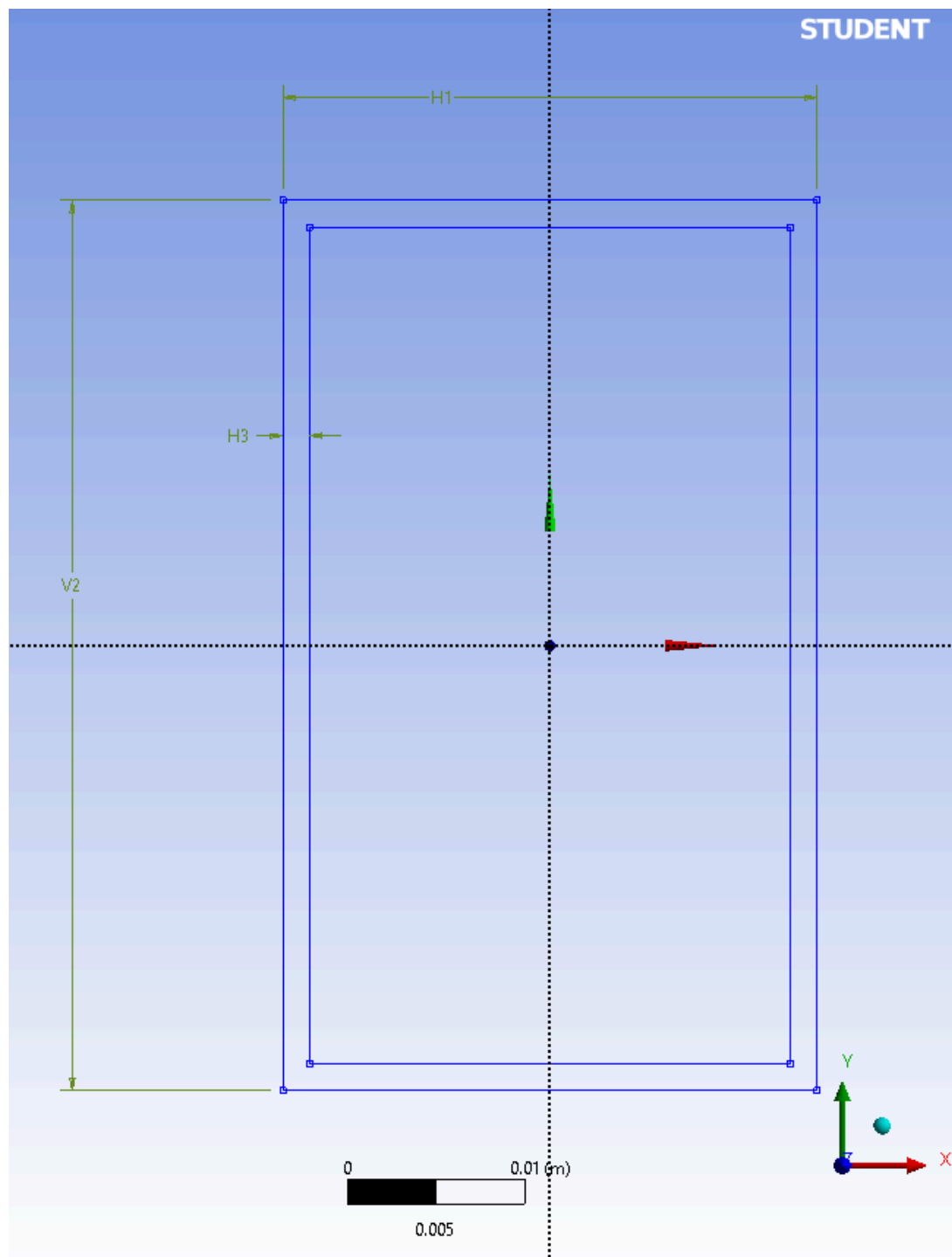


Figure 71: Top Tube.

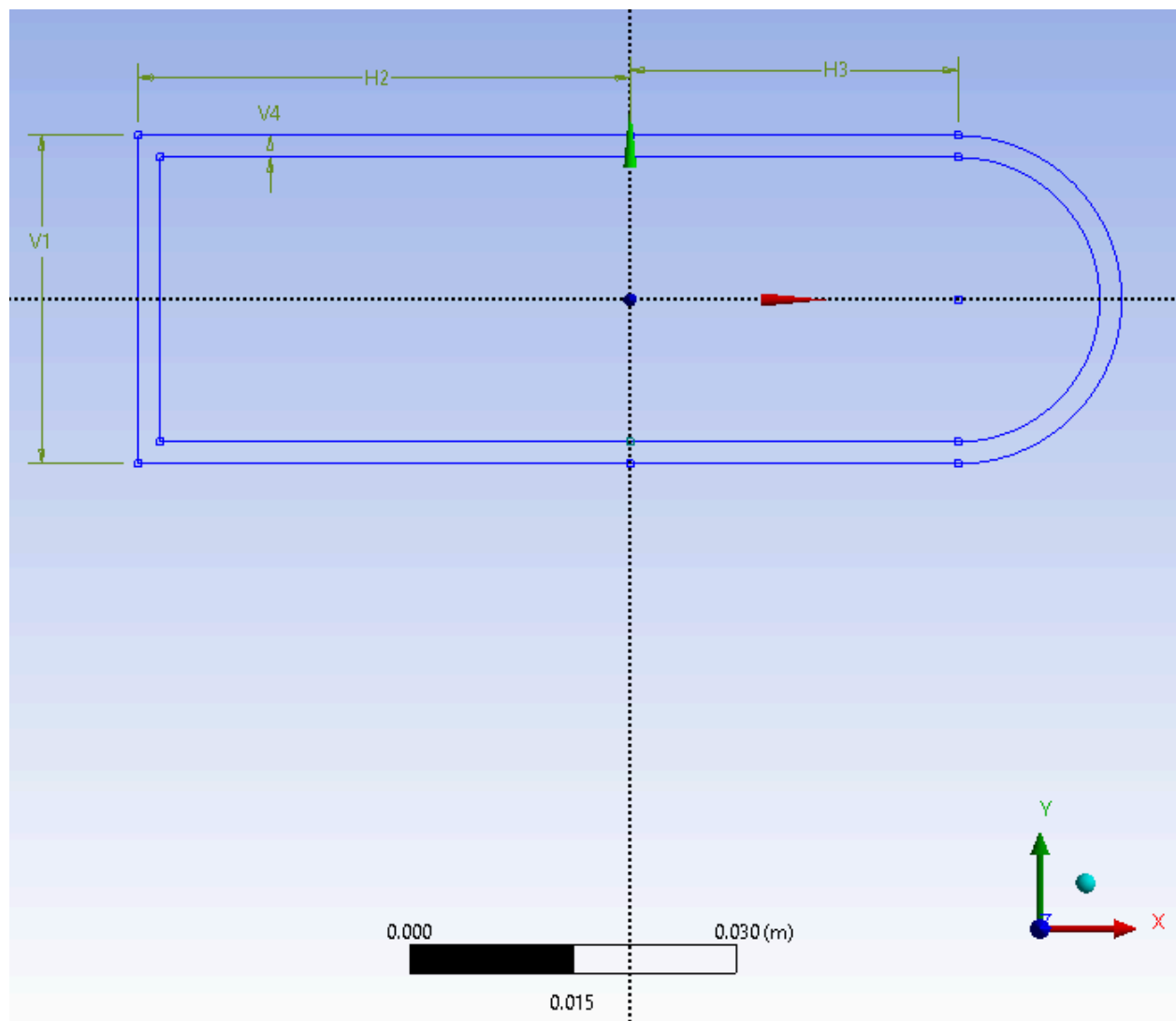


Figure 72: Front Tube.

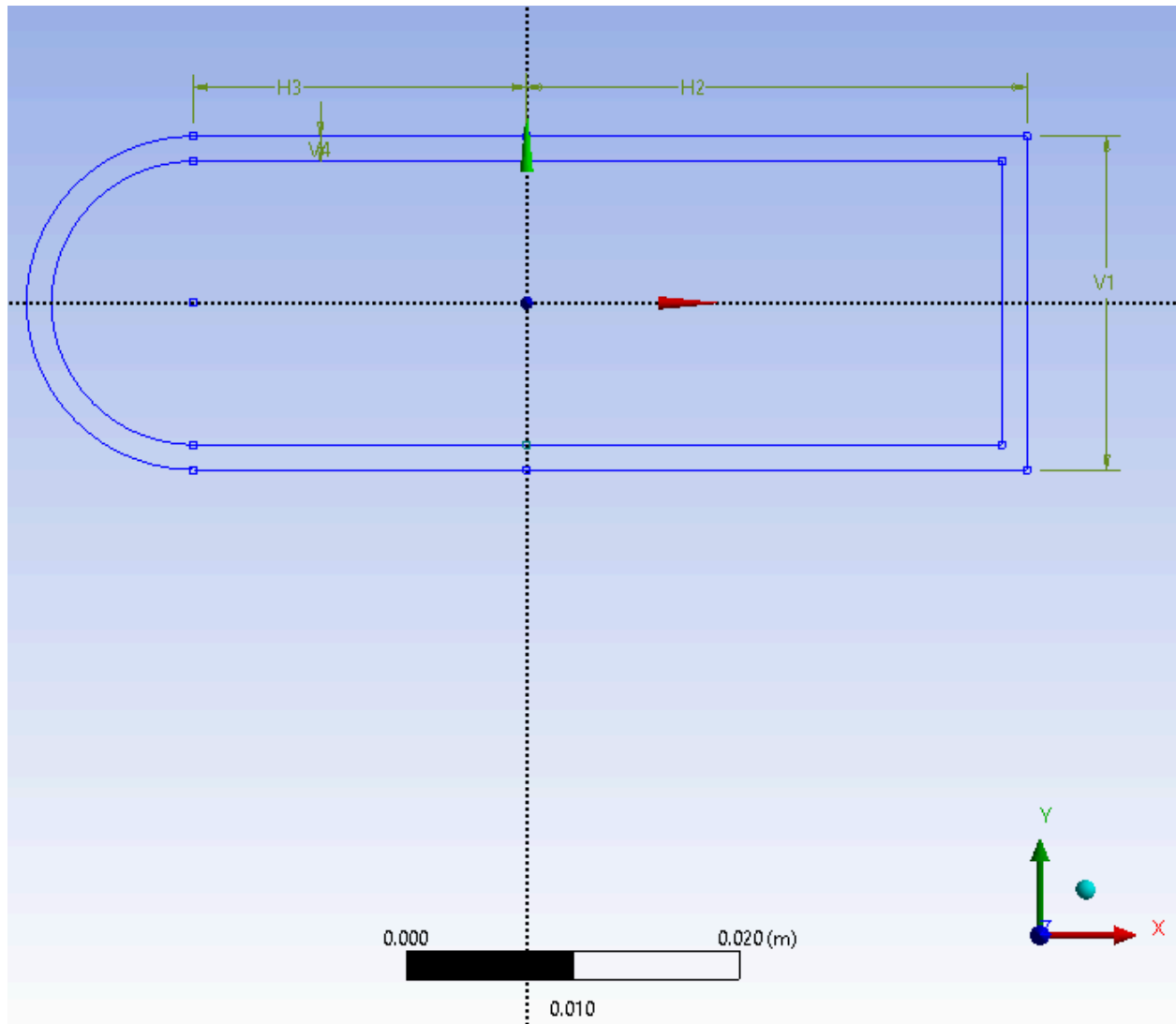


Figure 73: Seat Tube.

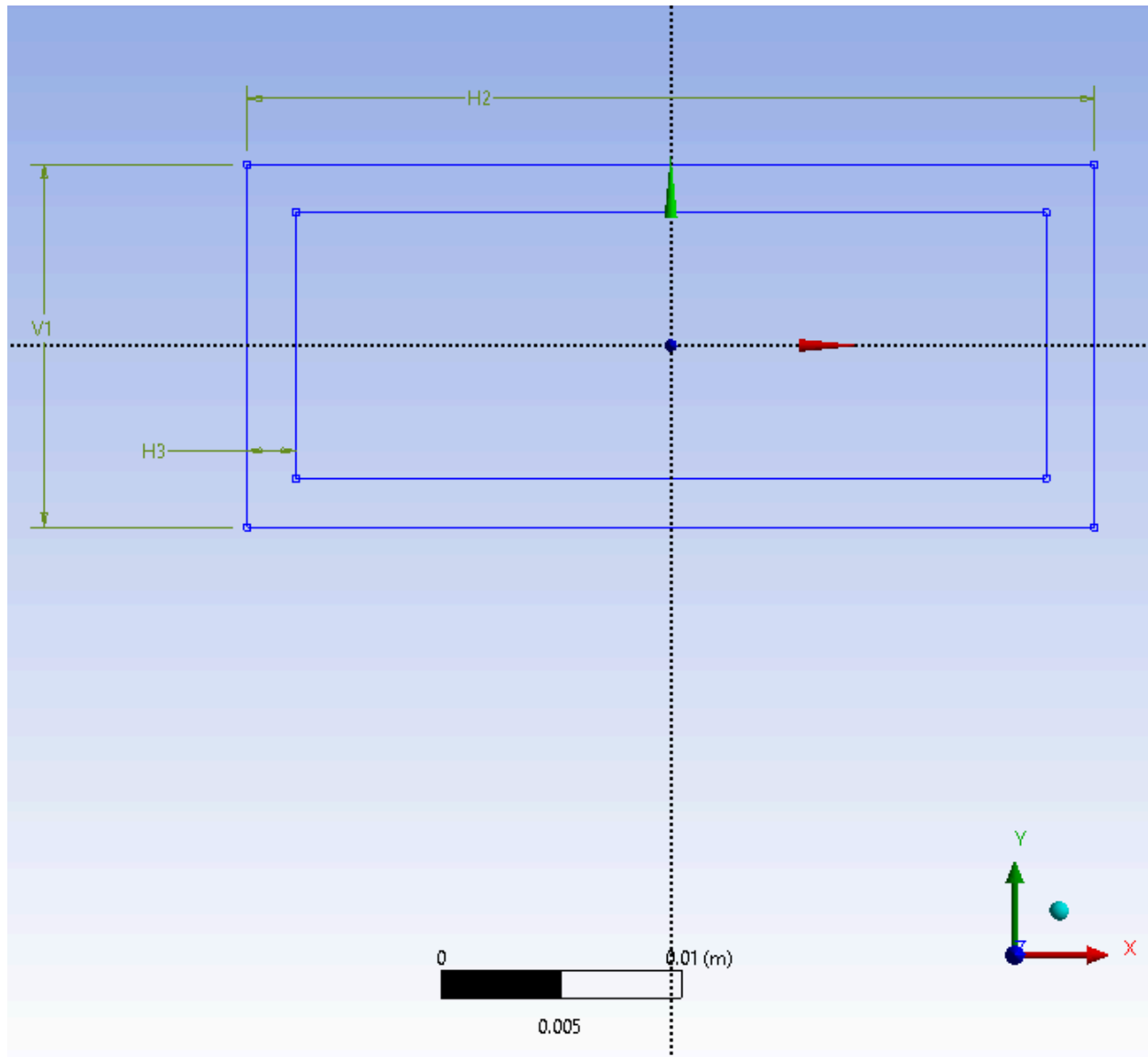


Figure 74: Stays Tube.